Which Adaptation Strategy for a New Task? A Controlled Comparison of Parameter-Efficient and Continual-Learning Strategies on 3D Brain MRI

Chi-Sheng Chen[1], Xinyu Zhang[2], En-Jui Kuo[3], Rong Fu[2], Qiuzhe Xie[4], and Fan Zhang[5]

***Abstract*—Adding a new analysis task to a deployed 3D brain-MRI model usually means a few dozen annotated cases, no access to earlier tasks' data, and a shared backbone. We compare eleven adaptation strategies, from LoRA, bottleneck adapters, bias-only tuning and linear probing to full fine-tuning, EWC, LwF and replay, on nine tasks from six public cohorts (four segmentation tasks; brain age, sex and dementia on IXI and OASIS-1) at 32 labelled volumes each, under one subject-grouped whole-volume protocol (five seeds, 676 runs, seed-level inference). What the new task inherits decides the strategy. Segmentation inherits a large decoder: training it on a frozen encoder is as accurate as LoRA on every task and better on one, and a frozen decoder with only its 66-parameter output projection trainable collapses. The global tasks inherit a $\sim$1K-parameter head, and their frozen features are more usable than gradient training suggests: a gradient-trained probe returns a constant, while the same probe fitted in closed form on standardised features matches LoRA on sex classification and comes within 0.7 yr of it on IXI brain age; moving the encoder pays only on the harder OASIS cohort. Full fine-tuning at its own learning rate is the most accurate single-task option, but a second task erases the first; EWC reduces the loss by a third without removing it, and only freezing shared weights gives zero forgetting. The pretrained checkpoint loads 59 of 82 segmentation tensors and is indistinguishable from random initialisation under LoRA on four tasks.**

***Index Terms*—Clinical model deployment, few-shot learning, continual learning, parameter-efficient fine-tuning, LoRA, 3D brain MRI.**

I. Introduction

A hospital imaging service rarely deploys one model and stops. A department running automated tumour segmentation may later want stroke-lesion delineation, then a quantitative marker such as brain age; each capability arrives with a small batch of newly annotated cases, months apart, and typically without access to the data behind the capabilities already in service, which is retained either not at all or under governance that makes reuse impractical [1], [2]. The engineering question this poses is concrete: given a frozen model that already works, 32 annotated volumes, and no prior-task data, which adaptation strategy should be used, and what does each cost to run and to store?

The options span four orders of magnitude of trainable parameters. At one end, full fine-tuning updates everything and is the usual default; it also overwrites the representation the earlier tasks depend on, causing *catastrophic forgetting* [3], [4], and requires a separate copy of the whole network per task. Regularisation- and distillation-based remedies such as EWC [3] and LwF [5] aim to soften that at the cost of tuning effort. At the other end, the backbone can be *frozen* and only small task-specific modules trained: a linear probe over frozen features, bias-only tuning [6], or low-rank adaptation (LoRA) [7], [8], which injects trainable low-rank matrices while leaving the original weights untouched. Freezing has an operational attraction beyond compute: because prior-task parameters are never modified, prior-task behaviour cannot change, and because nothing is replayed, no earlier patient data need be retained, which matters where retention is governed by HIPAA or GDPR.

Which of these to choose is usually settled by habit rather than evidence, and the evidence that exists is fragmented across single tasks and inconsistent protocols. We therefore ran the comparison directly: eleven strategies, nine tasks spanning segmentation, regression and classification, six public cohorts, five seeds, one protocol, 676 runs in all, 54 of them closed-form probe fits.

The answer is neither "adapters are efficient" nor "fine-tuning is better". It depends on what the new task inherits. Where it inherits a large task-specific head, that head is what has to be trained and the encoder can stay frozen. Where it inherits only a small head, the frozen features go further than gradient training suggests, and what the encoder adds depends on the cohort. Reaching that answer also required two corrections to our own first analysis, a per-strategy learning rate and statistics at the level of the training run, both of which narrowed claims we had made.

Reaching that answer required first settling how it is measured. The evaluation pipeline we initially built, following what is common in this literature, reported on the same split used for model selection, scored a single random 64^3 crop per case, pooled voxels across cases into one Dice, and let longitudinal timepoints of one patient span training and test. Each of these inflates results, and together they were enough to reverse conclusions: under that pipeline LoRA appeared to underperform linear probing on segmentation by 0.19 Dice, where

(Corresponding author: En-Jui Kuo.)
[1]Beth Israel Deaconess Medical Center & Harvard Medical School, Boston, MA 02115 USA.
[2]Luddy School of Informatics, Computing, and Engineering, Indiana University Bloomington, Bloomington, IN 47405 USA.
[3]Department of Electrophysics, National Yang Ming Chiao Tung University, Hsinchu 30010, Taiwan (e-mail: kuoenjui@nycu.edu.tw).
[4]Institute of Electronics Engineering, National Taiwan University, 106319, Taiwan.
[5]Department of Mathematics, Boise State University, Boise, ID 83702 USA.

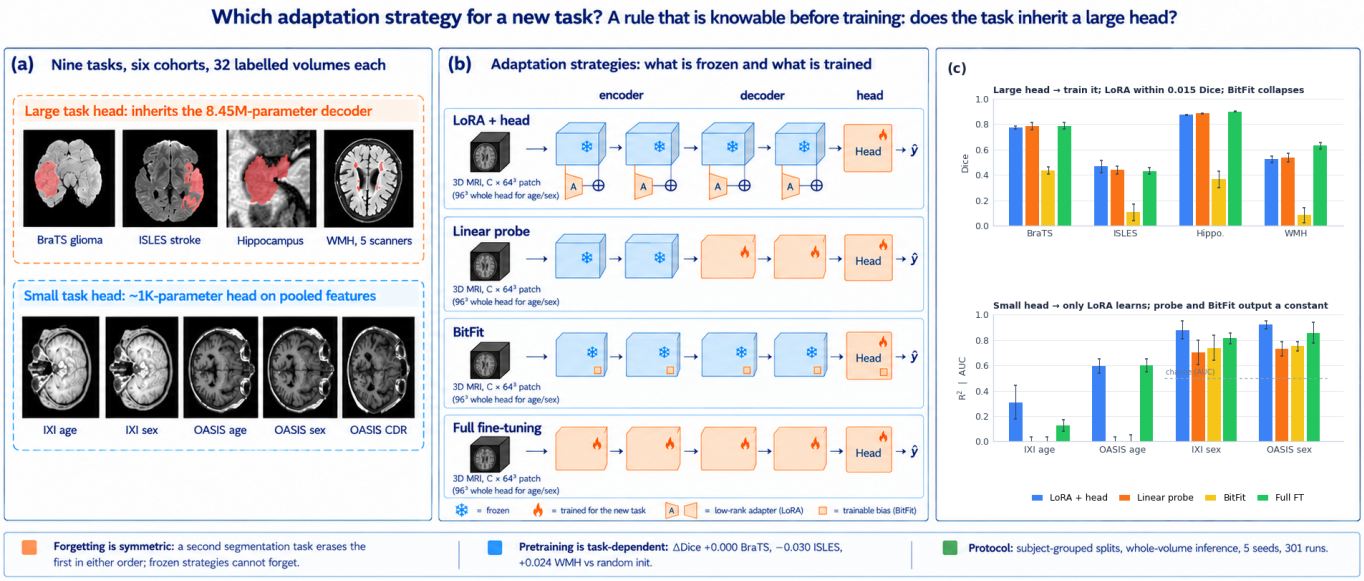


Fig. 1. Overview. (a) Nine brain-MRI tasks from six cohorts, each learned from 32 labelled volumes, grouped by whether the task inherits a large task-specific head (the U-Net decoder) or only a ∼1K-parameter head over pooled encoder features; one held-out test slice per task, segmentation targets in red. (b) Four of the eleven strategies compared, the ones that define the decision rule, drawn as what each freezes (snowflake) and trains (flame) in the same frozen FOMO 3D U-Net: LoRA + head adds low-rank branches to every convolution and trains the output projection; linear probing trains everything downstream of the encoder; BitFit trains bias terms and the head; full fine-tuning trains all 22.6 M parameters. The 66-parameter output projection is trained in every strategy; leaving it frozen is the single largest source of error we observed (Sec. V-B). (c) The four strategies at $N_k = 32$, mean ± SD over five seeds, at the shared learning rate: with a large head, training it is at least as accurate as LoRA and BitFit collapses; with a small head, LoRA learns while the gradient-trained linear probe and BitFit return a constant output, a failure that Sec. V-F traces to the optimiser rather than the features. Footer: the three further findings of the paper. Seed-level paired tests for every contrast are in Tables III and IV.

the true gap is 0.011. We replaced it with subject-grouped train/validation/test splits, whole-volume sliding-window inference, and per-case Dice and 95th-percentile Hausdorff distance (HD95), and report every number below under that protocol. Sec. IV-B specifies it; we release it with the code because the same failure modes are easy to reproduce accidentally.

Our findings and contributions are:

- **What the new task inherits decides the strategy, and the two halves of that rule rest on different evidence.** Segmentation inherits an 8.45M-parameter decoder; training it on a frozen encoder matches LoRA on BraTS, ISLES and WMH and beats it by 0.011 Dice on hippocampus (seed-level paired tests, Holm-corrected), and a frozen encoder with only the 66-parameter output projection trainable collapses on all four tasks (Dice 0.01–0.23), so it is the decoder that must be trained. The global tasks inherit a ∼1K-parameter head, and what fails there is the optimiser, not the features (next item).
- **"Frozen features cannot learn brain age or sex" was an optimisation artefact.** A gradient-trained linear probe returns a constant on every whole-head task, as does bias-only tuning on nearly every seed. A closed-form ridge or logistic probe on the same frozen features, standardised, learns all four: IXI sex AUC 0.878, equal to LoRA; IXI brain age 11.70 yr against LoRA's 10.99 ($p = 0.13$); OASIS brain age 15.46 against 11.95 and OASIS sex 0.845 against 0.920, where LoRA's lead is real. A 131K-parameter MLP head trained by gradient descent stays constant, and even with all 249 IXI subjects the gradient probe stays at 14.6 yr where the closed-form one reaches 9.1. Moving the encoder, by LoRA, bottleneck adapters or partial fine-tuning, helps on the harder cohort and is otherwise within noise of a well-fitted probe.
- **A learning rate shared across strategies favours the parameter-efficient ones.** At the shared rate of 10^{-3} full fine-tuning leads LoRA by 0.015 Dice on BraTS (not significant) and trails it by 0.06 AUC on IXI sex; with its rate chosen on validation (10^{-4}) it leads by 0.06 Dice and 0.03 AUC, while LoRA keeps the top of the grid on two of the three tasks. Full fine-tuning is the most accurate single-task option when tuned; the cost is forgetting.
- **Forgetting-free is a property of freezing, and the remedies only soften forgetting.** Frozen-backbone strategies have backward transfer exactly zero. Under full fine-tuning a second segmentation task erases the first in either order (-0.58 and -0.39 Dice), and on a two-task sequence whose second task is learnable (BraTS, then whole-head brain age) fine-tuning loses 0.54 Dice; EWC at $\lambda = 10^2$ reduces that to 0.38 ($p = 0.006$, the only remedy that survives correction) at a 0.8 yr cost on the new task, and LwF and replay, in output- and label-based forms as well as the feature-based ones, are indistinguishable from doing nothing.
- **The pretrained checkpoint should be validated, not assumed.** It fits 59 of the 82 segmentation tensors (the input convolution, one convolution per decoder stage and the output projection have the wrong shape; the second convolution of each decoder stage is absent). LoRA on a randomly initialised backbone reaches the same Dice

on BraTS, ISLES and hippocampus and the same brain-age MAE on IXI; the checkpoint helps by 0.024 Dice on WMH, 1.2 yr on OASIS brain age and 0.03–0.04 AUC on sex, and its frozen features beat random ones under the closed-form probe on every global task (1.5–3.0 yr, 0.08–0.21 AUC).

- **A 66-parameter configuration detail dominates.** Leaving the segmentation output projection frozen, as "frozen backbone" naively implies, costs 0.247 Dice on every seed and quadruples the seed SD.
- **A corrected evaluation protocol, seed-level statistics and code**, with tests that guard the label-handling and split-leakage failures we encountered, including one that silently zeroed every mask of a single-channel dataset while validation reported a perfect score.

II. Related Work

A. Foundation Models for Medical Imaging

Large-scale pretrained models have transformed medical image analysis. The FOMO initiative [1] provides 3D foundation models pretrained on diverse medical imaging datasets via masked autoencoding objectives, yielding general-purpose representations for downstream adaptation. UNETR [9] introduced Vision Transformers into 3D medical segmentation by replacing CNN encoders with self-attention, demonstrating strong performance on multi-organ and brain tumour segmentation. More recently, SAM-Med3D [10] extended the Segment Anything paradigm to volumetric data, and domain-specific foundation models have emerged for pathology [11] and radiology. These models share a common deployment challenge: adapting to multiple downstream tasks efficiently while preserving the pretrained representations, particularly when labelled data is scarce.

B. Continual Learning in Medical Imaging

Continual learning addresses the problem of sequentially learning new tasks without forgetting previously acquired knowledge [4]. Classical approaches fall into three categories: (i) *regularisation-based* methods such as EWC [3] and Synaptic Intelligence [12], which penalise changes to important parameters; (ii) *replay-based* methods such as iCaRL [13], which store or generate exemplars from prior tasks; and (iii) *distillation-based* methods such as LwF [5], which use the previous model's outputs as soft targets. In medical imaging, continual learning has been explored primarily for class-incremental segmentation: LifeLonger [14] benchmarks continual disease classification across dermatology datasets, while Lifelong nnU-Net [15] standardises continual organ segmentation. However, most prior work considers *class-incremental* settings with homogeneous outputs (e.g., adding new organ classes). Our work addresses *task-incremental* continual learning across *heterogeneous* output types (segmentation, regression and classification), which presents distinct challenges since the task heads and loss functions differ fundamentally, and, as Sec. V-E shows, the right adaptation strategy differs with them.

C. Parameter-Efficient Fine-Tuning

Parameter-efficient fine-tuning (PEFT) methods adapt large pretrained models by training only a small subset of parameters. LoRA [7] injects trainable low-rank decompositions into weight matrices, achieving competitive performance with full fine-tuning at a fraction of the parameter cost. Alternative PEFT approaches include Adapters [16], which insert bottleneck modules between layers; Prefix-Tuning [17], which prepends learnable tokens to the input; and BitFit [6], which tunes only bias terms. In the context of continual learning, PEFT methods offer a natural mechanism for task isolation: PackNet [18] iteratively prunes and freezes network capacity per task, while Piggyback [19] learns binary masks over a shared backbone. Of these, LoRA and bottleneck adapters apply directly to 3D convolutions, which is why they are the two PEFT families this comparison includes; neither is new here, and the contribution is the controlled comparison between them and the non-PEFT alternatives, not a method. A recurring detail in the PEFT literature, and the one that turns out to matter most here, is that the task-specific output layer is always trained alongside the low-rank updates; a pretrained projection cannot be expected to map onto a new label space.

III. Problem Setting and Strategies

A. Problem Formulation

We consider adding tasks to a shared backbone, in a sequence for the continual-learning experiments and one at a time for the single-task evaluations. Let f_θ denote a pretrained 3D U-Net backbone (frozen) and $\mathcal{T}_1, \mathcal{T}_2, \ldots, \mathcal{T}_K$ a sequence of tasks. For each task k, we have access to only N_k labelled few-shot samples ($N_k = 32$ throughout) drawn from task-specific distribution $\mathcal{D}_k = \{(x_i^{(k)}, y_i^{(k)})\}_{i=1}^{N_k}$. We assume *no replay*: when training on $\mathcal{T}_k$, we cannot access data from $\mathcal{T}_{<k}$. The optimisation objective for task k is:

$$\min_{\phi_k, h_k} \frac{1}{N_k} \sum_{i=1}^{N_k} \mathcal{L}_k\big(h_k(f_\theta(x_i^{(k)}; \phi_k)),\ y_i^{(k)}\big) \tag{1}$$

where ϕ_k denotes the task-specific trainable parameters inside the backbone (a LoRA adapter, a bottleneck adapter, bias terms, or nothing, depending on the strategy), h_k the task-specific head, and $\mathcal{L}_k$ the task loss. For the frozen-backbone strategies θ never changes; the fine-tuning baselines relax this and update θ as well. The goal is to learn $\{\phi_k, h_k\}_{k=1}^K$ such that performance on all seen tasks remains high after training on $\mathcal{T}_K$.

Tasks are of three output types, a 3D binary mask (segmentation), a scalar (regression) or a class (classification), each with its own modality set (one to three input channels) and its own output head h_k. Because the modality counts differ, the first convolution of the encoder cannot be shared and is task-specific; all remaining encoder weights (39 of 40 tensors) are shared and frozen.

Figure 1 gives the overview: the nine tasks grouped by head size (a), four of the strategies drawn as what each freezes and trains in the shared backbone (b), and the decision rule the results support (c). Under adapter isolation, inference for task

k uses backbone f_θ + adapter ϕ_k + head h_k; prior adapters and the backbone remain frozen. Task identity is assumed known at inference.

B. LoRA for 3D Convolutions

Low-Rank Adaptation (LoRA) [7] injects trainable low-rank matrices into a frozen weight matrix $W \in \mathbb{R}^{d\times k}$:

$$W' = W + \Delta W, \quad \Delta W = \frac{\alpha}{r} B \cdot A, \quad A \in \mathbb{R}^{r\times k},\ B \in \mathbb{R}^{d\times r} \tag{2}$$

with rank $r \ll \min(d, k)$ and scaling factor α. Only A and B are trained; W stays frozen. The scaling ratio α/r controls the magnitude of the low-rank perturbation relative to the pretrained weights.

For 3D convolutions in a U-Net, we apply LoRA via parallel $1 \times 1 \times 1$ convolutions acting on the channel dimension:

$$\text{Conv3d}'(x) = \text{Conv3d}(x; W) + \frac{\alpha}{r}\,\text{Conv1}(x; B) \circ \text{Conv1}(x; A) \tag{3}$$

where Conv1 denotes $1 \times 1 \times 1$ convolution, $A \in \mathbb{R}^{r\times C_{\text{in}}}$ projects to rank-r space, and $B \in \mathbb{R}^{C_{\text{out}}\times r}$ projects back. This formulation is memory-efficient and compatible with standard Conv3d layers [20].

We use $\alpha = 16$, evaluate $r \in \{4, 8, 16\}$ (Sec. V-C) and report $r = 16$, and target every Conv3d in both encoder and decoder for segmentation; encoder-only suffices for regression. For a single Conv3d layer with input channels C_{in} and output channels C_{out}, the number of LoRA parameters is:

$$|\phi_{\text{layer}}| = r \cdot (C_{\text{in}} + C_{\text{out}}) \tag{4}$$

which is substantially smaller than the original $C_{\text{out}} \times C_{\text{in}} \times D \times H \times W$ parameters. Summed over all targeted layers at $r = 16$, the adapter for segmentation has 93,810 parameters against a 22.58M-parameter U-Net (0.41%), and the adapter for the global tasks has 55,328 parameters (55,841 with the regression head) against the 14.14M-parameter encoder-plus-head (0.39%); at $r = 8$ the counts are 46,872 and 28,177.

C. Task Isolation by Freezing

Under every frozen-backbone strategy each task k receives its own trainable set ϕ_k (a LoRA adapter in Algorithm 1; a bottleneck adapter, bias terms or only the head in the other strategies). The trainable parameter set for task k is strictly disjoint from all other tasks:

$$\Phi_k = \{\phi_k, h_k\}, \quad \Phi_i \cap \Phi_j = \emptyset \ \ \forall\, i \neq j \tag{5}$$

Because θ and $\{\Phi_{<k}\}$ are frozen when optimising Φ_k, the performance of any prior task $j < k$ is invariant to training on $\mathcal{T}_k$. Formally, let $R_{t,j}$ denote the performance of task j after training up to task t. Then:

$$R_{t,j} = R_{j,j} \quad \forall\, t \geq j \implies \text{BWT}_j = R_{K,j} - R_{j,j} = 0 \tag{6}$$

This zero-forgetting property holds *by construction*, without regularisation or replay. It is not, however, unique to adapters: any scheme that leaves shared weights untouched, including linear probing on a frozen encoder, satisfies Eq. (6) equally. What distinguishes the schemes is how many parameters they must train to reach a given accuracy, which is what Sec. V measures. The total model storage after K tasks is:

$$\text{Storage} = |\theta| + \sum_{k=1}^{K} (|\phi_k| + |h_k|) \tag{7}$$

which grows linearly in K with a small per-task overhead since $|\phi_k| \ll |\theta|$.

Algorithm 1 Task-isolated adaptation of a frozen backbone (shown for LoRA)

Require: Pretrained backbone f_θ; task sequence $\mathcal{T}_1, \ldots, \mathcal{T}_K$; subject-grouped train/val/test splits per task; LoRA rank r; epochs E
Ensure: Trained adapters $\{\phi_k\}_{k=1}^K$ and heads $\{h_k\}_{k=1}^K$

```
Freeze all parameters of f_θ
for k = 1 to K do
    Initialise LoRA adapter φ_k with rank r (Kaiming A, zero B)
    Initialise task head h_k (output projection for segmentation; pooled regressor for regression); mark trainable
    if k > 1 then
        Freeze all previous adapters φ_1, ..., φ_{k-1} and heads h_1, ..., h_{k-1}
    end if
    for epoch = 1 to E do
        for each mini-batch (x, y) ∈ D_k^train do
            ŷ ← h_k(f_θ(x; φ_k))
            Update φ_k, h_k on L_k(ŷ, y) with AdamW
        end for
        Score D_k^val (deterministic centre crop); keep the best epoch
    end for
    Score the kept checkpoint once on D_k^test with whole-volume inference
end for
```

D. Task Heads and Losses

Segmentation. The full U-Net decoder produces a voxel-wise two-class prediction. LoRA is applied to every Conv3d in encoder and decoder. The head h_1 is the decoder's final $1\times1\times1$ projection from 32 channels to the 2 output classes (66 parameters), which is trained from its pretrained initialisation. We stress this because a natural but wrong reading of "frozen backbone" is to freeze this layer too; Sec. V-B quantifies the consequence. The training loss is cross-entropy plus soft Dice on the foreground probability:

$$\mathcal{L}_{\text{seg}} = \text{CE}(p, g) + 1 - \frac{2\sum_i p_i g_i + \epsilon}{\sum_i p_i + \sum_i g_i + \epsilon} \tag{8}$$

where p_i is the softmax foreground probability at voxel i, g_i the binary label, and $\epsilon = 10^{-8}$.

Global tasks (regression and classification). For brain age and the classification tasks we use the encoder only: its deepest 512-channel feature map is global-average-pooled and passed through a single linear layer, 513 parameters for a scalar

output and 1,026 for two logits. LoRA targets the encoder's Conv3d layers. The regression loss is mean squared error in years; classification uses cross-entropy, and per-case decisions take the higher logit (a 0.5 threshold on the positive-class probability).

Optimisation. All methods use AdamW (learning rate 10^{-3}, weight decay 10^{-5}) with cosine annealing to 10^{-6} over 100 epochs, batch size 2, gradient-norm clipping at 1.0, and LoRA dropout 0.1. Training patches are random 64^3 crops; no other augmentation is used. Hyperparameters are identical across methods and were not tuned per method.

IV. Experiments

A. Datasets

Notation. We use nine tasks from six cohorts: T1 glioma segmentation, T2 brain-age regression and T3 stroke-lesion segmentation form the continual sequences (Sec. V-I); T4 hippocampus segmentation, T5 sex classification and T6 white-matter-hyperintensity segmentation are single-task evaluations that test whether the two patterns we find replicate on segmentation and global tasks drawn from outside the original pair; and T7 brain age, T8 sex and T9 dementia classification repeat the global tasks on a second cohort (OASIS-1) acquired on a different scanner.

T1: BraTS 2023 Glioma [21], [22]. We use the 1251 labelled volumes of the training release, with channels T2w, T2-FLAIR and T1c and a binary whole-tumour mask derived from the expert annotation. The challenge's validation release ships without segmentations and is excluded; an earlier pipeline had written those 219 volumes with all-zero masks, which are indistinguishable from a tumour-free label and had contaminated 17% of the test set. Several subjects contribute more than one longitudinal timepoint; these are kept in the same split.

T2: IXI [23]. Of the 577 subjects with both T1 and T2 scans, 559 have a usable age in the IXI demographics table (15 have no table entry, 3 have a missing age) and are used; none is imputed. Test-set ages span 20.0–86.3 yr (mean 50.0, SD 16.5).

T3: ISLES 2022 [24]. Ischemic-stroke lesion segmentation with channels DWI, ADC and FLAIR (FLAIR resampled onto the DWI grid) and the expert lesion mask; 247 of the 250 cases are used, three being excluded because their mask emptied under 1 mm resampling. Lesions are small (median $\approx$775 voxels), which makes this a harder segmentation task than T1 and one drawn from a single cohort, so unlike an earlier infarct-*classification* design that contrasted ISLES with IXI controls, it carries no acquisition confound.

T2 (whole-head variant). Because patch input proved to be the limiting factor for regression (Sec. V-D), we also prepare each IXI subject as a single 96^3 cube: the volume is cropped to its non-zero content and resampled, so one forward pass sees the entire head. The task, labels, splits and metrics are otherwise identical, and models are run with a 96^3 window, which makes the crop the identity and evaluation a single window.

T4: Medical Segmentation Decathlon (MSD) Task04 hippocampus [25]. 260 single-channel T1 volumes with anterior and posterior hippocampus labels, merged to binary. This task shares none of the incidental properties of T1 and T3: one input channel rather than three, healthy anatomy rather than pathology, a large consistent foreground (4.7% of voxels against $<$0.1% for a lesion), and volumes of roughly $35\times51\times35$ that fit inside the 64^3 window, so the whole structure is visible in one forward pass. It therefore tests whether the segmentation result is about the presence of a large decoder or about tumours.

T5: IXI sex classification. The same 559 subjects, scans, modalities and whole-head 96^3 cubes as T2, labelled instead by the sex recorded in the IXI demographics table (247 male, 312 female; majority-class accuracy 0.56). It differs from T2 only in output type, which makes it a direct test of whether the regression finding is about the output or about where capacity must sit. It contributes the study's classification task; an infarct-detection formulation contrasting two cohorts was rejected as confounded by acquisition (Sec. VI-B).

T6: WMH Segmentation Challenge 2017 [26]. 170 subjects from three sites and five scanners (Utrecht 3T Philips; Singapore 3T Siemens; Amsterdam 3T GE, 1.5T GE and 3T Philips PET-MR), each with a FLAIR and a co-registered T1 and an expert white-matter-hyperintensity mask; the challenge's "other pathology" label is mapped to background, as in its own scoring. It is the only task whose test set spans several scanners, so it adds a multi-site acquisition shift to the segmentation family as well as a fourth pathology.

T7–T9: OASIS-1 [27]. 416 subjects aged 18–96 with a single 1.5T Siemens T1 (the challenge-supplied N4-corrected, 1 mm native-space average of the MPRAGE repeats), prepared as 96^3 whole-head cubes exactly as the IXI tasks are. T7 repeats brain-age regression and T8 sex classification (160 male, 256 female) on this second cohort; T9 labels the 198 subjects aged 60 and over by Clinical Dementia Rating, 100 with CDR$>$0 against 98 at CDR 0, a clinical classification that IXI cannot provide. The 20 reliability re-scans are dropped so every subject appears once.

Data use. BraTS 2023 was obtained from Synapse (syn51156910) under its Data Use Agreement; IXI is publicly available at https://brain-development.org/ixi-dataset/; ISLES 2022 is distributed under CC BY 4.0 (Zenodo 7153326); WMH 2017 under CC BY-NC 4.0 (DOI 10.34894/AECRSD); OASIS-1 under its Data Use Agreement (https://sites.wustl.edu/oasisbrains/). No additional ethics approval was required.

Preprocessing. Volumes are resampled to 1 mm isotropic and z-scored per channel over non-zero voxels.

B. Splits and Evaluation Protocol

For each task and seed we draw a subject-grouped three-way split: a training set of exactly $N_k = 32$ volumes, a validation set of 60 subjects used only for epoch selection, and a held-out test set of up to 250 subjects (268–277 volumes for T1 depending on the seed; 250 for T2, T5, T7 and T8; all remaining subjects elsewhere: 155 for T3, 168 for T4, 78 for T6, 106 for T9) on which every number in this paper is

reported. No subject appears in more than one split. Splits are persisted so that all methods are scored on identical cases, which is what makes the paired comparisons below valid.

Segmentation is scored on the whole volume by Gaussian-weighted sliding-window inference (64^3 windows, 25% overlap), followed by argmax. We report per-case Dice and per-case HD95 in voxel units, averaged over cases. **Regression** is scored by averaging the prediction over a fixed $2\times2\times2$ grid of 64^3 windows. We report MAE in years, R^2, and the standard deviation of the predictions, the last because it exposes a model that has collapsed to a constant regardless of MAE.

Backward transfer. For the sequential baselines, after each later task we re-score every earlier task's test set with that task's own head on the now-updated encoder and report $\mathrm{BWT}_j = R_{K,j} - R_{j,j}$ in Dice; in the three-task sequence we also report the intermediate value after T3. For adapter-isolated methods this quantity is zero by construction and is not re-measured. Both sequences are also run in reverse order (T2→T1 and T2→T3→T1) to test whether forgetting depends on which task comes first.

Statistics. Means and SDs are over seeds, and the training run is the unit of inference. For each contrast the per-case differences are averaged within a seed (all methods share the split within a seed; for classification the per-seed AUC difference is used), and the five per-seed values give the mean, a t-based 95% CI, a paired t-test against zero and the number of seeds favouring the first method. Within each table of contrasts, p-values are Holm-adjusted for the number of contrasts in that table and both raw and adjusted values are shown. Per-case Wilcoxon and McNemar statistics, which pool recurring subjects across seeds, are released as descriptive numbers only.

C. Backbone

`unet_b` from the FOMO (Foundation Models for Open Medical imaging) challenge [1], initialised from `fomo25_mmunetvae_pretrained.ckpt`. The checkpoint supplies 59 of the segmentation model's 82 tensors: 39 of the 40 encoder tensors (the modality-specific input convolution is re-initialised) and 20 of the 42 decoder tensors; the first convolution of each decoder stage and the output projection have task-dependent shapes and are re-initialised, and the second convolution of each decoder stage is absent from the checkpoint (Sec. V-K). The same checkpoint is loaded at the start of every task for every method.

D. Compared Methods

All methods share the backbone, data, splits, optimiser and schedule; they differ only in which parameters are trainable (Table I).

- **LoRA (enc+dec).** Rank-8 LoRA on every Conv3d with the output projection left frozen, the configuration a literal reading of "freeze the backbone" produces. Included to quantify that reading (Sec. V-B).
- **LoRA + head.** As above, plus the 66-parameter output projection. Reported at rank 16; ranks 4 and 8 appear in the ablation of Sec. V-C.
- **Sequential Linear.** Encoder frozen; every downstream parameter trainable. For segmentation this is the entire decoder (8.45M parameters); for regression, the 513-parameter head. One model carried through the sequence.
- **Sequential FT.** All parameters trainable; one model carried through the sequence. The forgetting reference.
- **BitFit** [6]. PEFT comparator below LoRA's budget: bias terms plus the output projection (6.4K parameters), task-isolated like LoRA.
- **Bottleneck adapters** [16]. PEFT comparator at LoRA's budget. Where LoRA adds a low-rank *parallel* update to a frozen convolution, a Houlsby adapter adds a small *serial* bottleneck after it: $y \leftarrow y + U\,\mathrm{GELU}(D\,y)$ with D a $1\times1\times1$ projection to 16 channels and U back, U zero-initialised so training starts from the frozen network's function. Inserted after every $3\times3\times3$ convolution LoRA targets, plus the output projection: 97.5K parameters for segmentation and 66.1K for brain age (LoRA r=16: 93.8K and 55.8K, each count including the 513-parameter head), task-isolated like LoRA.
- **EWC** [3], **LwF** [5], **experience replay**. Full fine-tuning plus a forgetting-mitigation term: a diagonal-Fisher penalty ($\lambda = 10^3$), encoder-feature distillation from the previous task's model (weight 0.5), and the same distillation on a 100-sample buffer of previous-task volumes, respectively.
- **Random-initialisation control.** LoRA + head (r=16) with the backbone initialised randomly instead of from the checkpoint, to test whether pretraining contributes anything (Sec. V-K); Sequential Linear and Sequential FT are also run both ways on BraTS and IXI brain age.
- **Confound controls** (Sec. V-F). *Head only*: encoder and decoder frozen, only the 66-parameter output projection trained (segmentation). *Closed-form probe*: the 513-parameter head fitted by ridge regression or L2-regularised logistic regression on z-scored frozen pooled features, with the penalty chosen on the validation split (global tasks; also on random features and on all labelled subjects). *MLP head*: a two-hidden-layer head (512 and 256 units, 131K parameters) trained by gradient descent on the same frozen features. *Partial fine-tuning*: the deepest encoder stage and everything downstream trainable. *Learning-rate sweep*: the five main strategies at 10^{-4}, 3×10^{-4} and 10^{-3} on BraTS, IXI brain age and IXI sex, seeds 42–44, rate chosen on validation.
- **Forgetting remedies on a learnable second task** (Sec. V-I). BraTS followed by whole-head IXI brain age, every run initialised from the same seed's BraTS checkpoint of Sequential FT: EWC at $\lambda \in \{10^2, 10^3, 10^4\}$, LwF as feature distillation and as output distillation of the previous BraTS head, and replay as feature distillation and as labelled rehearsal on a 100-volume buffer.
- **Full-data reference.** LoRA + head (r=16), Sequential Linear and Sequential FT trained on every labelled subject outside the seed-42 validation and test sets, one seed each: 908 BraTS volumes (10 epochs, about 2.8× the few-shot gradient budget), 249 IXI and 106 OASIS subjects (the 100-epoch recipe unchanged). The runs

TABLE I
TRAINABLE PARAMETERS PER METHOD, WITH THE SHARE OF THE RESULTING MODEL'S PARAMETERS (LORA AND ADAPTERS ADD TO THE 22,582,978-PARAMETER U-NET). GLOBAL-TASK COUNTS ARE FOR BRAIN AGE AND INCLUDE ITS 513-PARAMETER HEAD. THE COMPARISON IN TABLE II SPANS FOUR ORDERS OF MAGNITUDE OF TRAINABLE BUDGET.

	T1 segmentation		T2 regression	
Method	trainable	share	trainable	share
LoRA (enc+dec), r=8	46,872	0.207%	28,177	0.199%
LoRA + head, r=8	46,938	0.207%	28,177	0.199%
LoRA + head, r=16	93,810	0.414%	55,841	0.393%
Bottleneck adapters	97,506	0.430%	66,145	0.466%
Sequential Linear	8,446,306	37.4%	513	0.004%
Sequential FT	22,582,978	100%	14,136,321	100%

show how far the 32-shot results sit below what the same strategies reach with all the data (Sec. V-G). ISLES, hippocampus, WMH and CDR have no subjects outside the few-shot protocol's validation and test sets, so no such reference exists for them.

- **Second backbone.** LoRA + head (r=16), Sequential Linear, Sequential FT and BitFit repeated on a Swin UNETR [28] whose transformer encoder carries the self-supervised weights of Tang et al. [29] (feature size 48; 62.2M parameters for segmentation, 8.1M encoder-only), on BraTS, IXI brain age and IXI sex, seeds 42–44, with the training recipe unchanged. LoRA sits on every linear layer of the encoder (attention and MLP) and every $3\times3\times3$ convolution of the decoder; Sequential Linear trains the whole Swin UNETR decoder (54.1M) or the pooled head; BitFit trains every bias, LayerNorm offsets included (Sec. V-H).

Counting the three LoRA + head ranks separately, the first six items (with EWC, LwF and replay as three) are the eleven strategies compared throughout; the remaining items are controls and references. Trainable counts quoted for the global tasks include the head (513 parameters for regression, 1,026 for classification) unless stated otherwise. Per-task oracles (one independent model per task, no continual constraint) were run only under the superseded protocol and are omitted rather than reported on a different footing.

E. Implementation and Resources

Experiments run on a single NVIDIA RTX 3090. Measured peak GPU memory is 1.67 GB during segmentation training (batch 2) and 4.16 GB during whole-volume evaluation (batch 8 windows); regression peaks at 0.69 GB and 2.53 GB respectively. One LoRA segmentation run trains in 41.5 min (Table XVIII) and evaluates about 275 volumes in about 25 min; a whole-head regression run trains and evaluates in under 10 min. Code, splits and per-case predictions are released with the paper.

V. RESULTS

All results are on held-out test splits at $N_k = 32$, reported as mean ± SD over five seeds (42–46; the reversed-order sequences, the learning-rate sweep and the Swin UNETR replication use three); each seed draws its own subject-grouped split, and all methods share the split within a seed. The unit of inference is the training run. For every contrast the per-case differences are averaged within a seed (for classification, the per-seed AUC difference is used), and the five per-seed values give the mean, a t-based 95% CI, a paired t-test against zero, the number of seeds favouring the first method, and a Holm adjustment within the table the contrast appears in. Pooling the roughly 1,250 per-case differences of five runs, in which the same subjects recur in up to five test sets, would treat pseudo-replicates as independent; the per-case Wilcoxon statistics are kept in the released JSON as descriptive numbers only. With $n = 5$ a contrast needs to be consistent across seeds to register; several differences that the pooled tests had called significant do not survive this, and we say which. All 676 runs completed; six Swin BitFit runs first hit a PyTorch layer-norm kernel fault and were rerun with the affine step applied outside the fused kernel, which is numerically identical (Sec. V-H).

A. Main Comparison

Table II gives the comparison on the two-task sequence and Table III the paired tests.

B. The Output Projection Accounts for Most of the LoRA Gap

At rank 8, with the decoder's final $1\times1\times1$ projection frozen, LoRA on every other layer reaches Dice 0.503 ± 0.117 and HD95 135 ± 17; the spread across seeds reflects how much the outcome depends on whether the pretrained projection happens to be usable for the new label. Training those 66 additional parameters lifts it to 0.750 ± 0.027 and HD95 48.8 ± 4.6, a paired gain of $+0.247$ Dice on every seed (per-seed range $+0.09$ to $+0.39$, $p = 0.01$; Table III) at a budget that changes from 46,872 to 46,938, and a fourfold reduction in seed SD. Mechanically the pretrained projection maps decoder channels onto the pretraining objective's output space, and rank-8 perturbations upstream cannot re-map it onto a new label space. Every PEFT recipe trains the new head; a "frozen backbone" should not be read as including it.

C. Accuracy per Trainable Parameter, and Rank

Sequential Linear and LoRA + head at rank 16 are not distinguishable on BraTS: Dice 0.782 ± 0.032 vs. 0.772 ± 0.011, a paired -0.011 whose per-seed values run from -0.063 to $+0.044$ ($p = 0.57$), and HD95 43.5 vs. 44.8, with LoRA the more seed-stable of the two. Sequential Linear reaches its accuracy by training the full 8.45M-parameter decoder; LoRA + head trains 93.8K, a factor of 90 fewer. Full fine-tuning at the shared learning rate is $+0.015$ above LoRA + head ($p = 0.30$), at 240× the budget and at the cost of forgetting (Sec. V-I); with its own learning rate it is clearly better (Sec. V-F).

Table V varies the rank. Rank 4 is not worse than rank 8 (0.754 vs. 0.750) at half the parameters, so rank 8 sits in a flat region; rank 16 gains $+0.022$ over rank 8 on four of five seeds ($p = 0.09$) and cuts the seed SD from 0.027 to 0.011. Rank 16 is also the best of the three on the validation split

TABLE II

HELD-OUT TEST RESULTS, $N_k = 32$, MEAN $\pm$ SD OVER SEEDS 42–46, SEQUENCE T1$\rightarrow$T2. T1 = BRATS GLIOMA SEGMENTATION (250 SUBJECTS, 268–277 VOLUMES PER SEED); T2 = IXI BRAIN AGE FROM 64^3 PATCHES (250 SUBJECTS). "T1 AFTER T2" IS T1 RE-SCORED AFTER THE MODEL HAS LEARNED T2; BWT IS ITS CHANGE. A CONSTANT PREDICTOR OF THE TRAINING-MEAN AGE HAS MAE $\approx$14.7 YR ON T2.

	T1 segmentation		T2 brain age (patch)		Forgetting		
Method	Dice↑	HD95↓	MAE (yr)↓	R^2	T1 after T2	BWT	Trainable
Frozen backbone (task-isolated; BWT = 0 by construction)							
BitFit	0.436 ± 0.031	77.1 ± 9.3	14.66 ± 0.24	-0.03	= T1	0.000	6,434
LoRA (enc+dec), r=8	0.503 ± 0.117	135.1 ± 17.1	14.47 ± 0.35	-0.04	= T1	0.000	46,872
LoRA + head, r=4	0.754 ± 0.032	49.0 ± 6.6	14.32 ± 0.35	-0.00	= T1	0.000	23,502
LoRA + head, r=8	0.750 ± 0.027	48.8 ± 4.6	14.43 ± 0.27	-0.01	= T1	0.000	46,938
LoRA + head, r=16	0.772 ± 0.011	44.8 ± 2.4	14.67 ± 0.49	-0.05	= T1	0.000	93,810
Sequential Linear	0.782 ± 0.032	43.5 ± 7.1	14.71 ± 0.23	-0.04	0.782 ± 0.032	0.000	8,446,306
Backbone updated per task							
Sequential FT	0.787 ± 0.028	43.0 ± 6.5	14.65 ± 0.27	-0.02	0.455 ± 0.149	-0.331 ± 0.148	22,582,978
EWC	0.784 ± 0.027	43.7 ± 5.7	14.65 ± 0.23	-0.03	0.463 ± 0.157	-0.321 ± 0.181	22,582,978
LwF	0.789 ± 0.028	42.4 ± 6.4	14.69 ± 0.22	-0.03	0.315 ± 0.162	-0.475 ± 0.171	22,582,978
Replay	0.754 ± 0.043	50.1 ± 9.2	14.64 ± 0.30	-0.03	0.354 ± 0.141	-0.400 ± 0.103	22,582,978

TABLE III

PAIRED DIFFERENCES IN DICE ON THE FOUR SEGMENTATION TASKS WITH THE TRAINING RUN AS THE UNIT: MEAN OVER FIVE SEEDS OF THE WITHIN-SEED MEAN PER-CASE DIFFERENCE, t-BASED 95% CI, THE RANGE OF THE FIVE PER-SEED VALUES, PAIRED t-TEST p (RAW AND HOLM-ADJUSTED WITHIN THIS TABLE) AND THE NUMBER OF SEEDS FAVOURING THE FIRST METHOD. L+H$_r$ = LORA + HEAD AT RANK r; SL = SEQUENTIAL LINEAR; FT = SEQUENTIAL FT; L$_8$ = LORA (ENC+DEC) WITH THE OUTPUT PROJECTION FROZEN; RI = THE SAME LORA + HEAD CONFIGURATION ON A RANDOMLY INITIALISED BACKBONE; AD = BOTTLENECK ADAPTERS. ALL ISLES ROWS USE SINGLE-TASK RUNS; "SINGLE $-$ SEQ." COMPARES LORA TRAINED ON ISLES ALONE WITH LORA LEARNING ISLES SECOND IN THE THREE-TASK SEQUENCE.

Contrast	ΔDice	95% CI	per-seed range	p	p_{Holm}	seeds
BraTS: L+h$_{16}$ $-$ SL	-0.011	$[-0.057, +0.036]$	-0.063 to $+0.044$	0.57	1.00	1/5
BraTS: FT $-$ L+h$_{16}$	$+0.015$	$[-0.019, +0.049]$	-0.017 to $+0.043$	0.30	1.00	3/5
BraTS: L+h$_{16}$ $-$ L+h$_8$	$+0.022$	$[-0.005, +0.049]$	-0.011 to $+0.048$	0.09	1.00	4/5
BraTS: L+h$_8$ $-$ L$_8$	$+0.247$	$[+0.098, +0.397]$	$+0.094$ to $+0.394$	0.01	0.16	5/5
BraTS: L+h$_{16}$ $-$ RI	$+0.000$	$[-0.023, +0.023]$	-0.023 to $+0.029$	0.99	1.00	1/5
BraTS: L+h$_{16}$ $-$ AD	$+0.017$	$[-0.036, +0.070]$	-0.023 to $+0.088$	0.42	1.00	3/5
ISLES: SL $-$ L+h$_{16}$	-0.050	$[-0.149, +0.050]$	-0.137 to $+0.073$	0.24	1.00	1/5
ISLES: FT $-$ L+h$_{16}$	-0.057	$[-0.178, +0.064]$	-0.178 to $+0.070$	0.26	1.00	2/5
ISLES: L+h$_{16}$ $-$ RI	-0.030	$[-0.108, +0.049]$	-0.111 to $+0.062$	0.35	1.00	1/5
ISLES: L+h$_{16}$ $-$ AD	$+0.104$	$[+0.012, +0.196]$	$+0.004$ to $+0.179$	0.03	0.49	5/5
ISLES: L+h$_{16}$ single $-$ seq.	$+0.027$	$[-0.044, +0.098]$	-0.036 to $+0.097$	0.36	1.00	3/5
Hippocampus: SL $-$ L+h$_{16}$	$+0.011$	$[+0.007, +0.014]$	$+0.007$ to $+0.015$	1×10^{-3}	0.02	5/5
Hippocampus: FT $-$ L+h$_{16}$	$+0.025$	$[+0.022, +0.029]$	$+0.023$ to $+0.030$	5×10^{-5}	1×10^{-3}	5/5
Hippocampus: L+h$_{16}$ $-$ RI	$+0.002$	$[-0.000, +0.003]$	$+0.000$ to $+0.003$	0.06	0.75	5/5
Hippocampus: L+h$_{16}$ $-$ AD	-0.002	$[-0.004, -0.001]$	-0.004 to -0.001	0.02	0.35	0/5
WMH: SL $-$ L+h$_{16}$	$+0.015$	$[-0.052, +0.082]$	-0.057 to $+0.089$	0.58	1.00	3/5
WMH: FT $-$ L+h$_{16}$	$+0.108$	$[+0.081, +0.135]$	$+0.074$ to $+0.131$	4×10^{-4}	7×10^{-3}	5/5
WMH: L+h$_{16}$ $-$ RI	$+0.024$	$[-0.028, +0.076]$	-0.013 to $+0.096$	0.27	1.00	4/5
WMH: L+h$_{16}$ $-$ AD	-0.003	$[-0.050, +0.044]$	-0.031 to $+0.061$	0.87	1.00	1/5

(0.837 against 0.825 and 0.821), so its selection does not rest on the test set. BitFit, at 6.4K parameters, is far too small for this task (0.436) and marks the lower end of the budget curve.

The gap also closes with data (Table VI, Fig. 2): at 16 shots linear probing leads by 0.101, at 32 by 0.011, and at 64 the two are equal (0.816 both); with all 908 volumes they are 0.859 and 0.849 (Sec. V-G). Whatever price LoRA pays for its parameter count is paid in the smallest-data regime, the opposite of the intuition that adapters are the few-shot method.

D. Brain Age: Input Scope Decides, and LoRA Wins

From 64^3 patches, no method learns brain age: all ten configurations in Table II land at 14.3–14.7 yr, the level of a constant predictor (14.7 yr for the training mean, 14.5 for the test mean), with $R^2 \approx 0$, including full fine-tuning of every parameter. A 64^3 window covers a few percent of a $240\times240\times180$ head, and the cues for age (ventricular volume, cortical thinning, global atrophy) are not local.

Re-running the task with the whole head resampled to a single 96^3 cube changes the result (Table VII). LoRA + head reaches MAE 10.99 ± 1.27 yr with $R^2 = 0.31$, beating the constant predictor by 3.68 yr per subject on every seed ($p = 0.002$) and full fine-tuning at the shared learning rate by 1.68 yr ($p = 0.01$, $p_{\text{Holm}} = 0.06$; Table IV). The gradient-trained Sequential Linear head and BitFit do *not* recover: their predictions still have a standard deviation of about 0.1 yr against a true 16.5, i.e. they remain constant. Sec. V-F shows that this is a property of how the head is fitted, not of the frozen features.

Figure 3 shows what the numbers summarise: the linear probe's predictions sit in a 47–54 yr band whatever the true age, while LoRA's follow it, on both cohorts.

This result runs opposite to segmentation in one respect:

TABLE IV
PAIRED CONTRASTS ON THE GLOBAL TASKS, SEED AS THE UNIT (FIVE SEEDS; COLUMNS AS IN TABLE III). BRAIN AGE: WITHIN-SEED MEAN DIFFERENCE IN ABSOLUTE ERROR IN YEARS (NEGATIVE FAVOURS THE FIRST METHOD; "CONSTANT" PREDICTS THE MEAN TRAINING AGE). SEX AND CDR: DIFFERENCE OF PER-SEED AUCS (POSITIVE FAVOURS THE FIRST METHOD). p_{HOLM} IS ADJUSTED WITHIN EACH BLOCK.

Contrast	Δ	95% CI	range	p	p_{Holm}	seeds
Brain age, Δ|error| (yr)						
IXI: L+h$_{16}$ − constant	−3.68	[−5.14, −2.23]	−5.47 to −2.36	2×10^{-3}	0.02	0/5
IXI: L+h$_{16}$ − FT	−1.68	[−2.70, −0.67]	−2.86 to −0.82	1×10^{-2}	0.06	0/5
IXI: L+h$_{16}$ − RI	−0.00	[−1.73, +1.73]	−1.70 to +2.00	1.00	1.00	3/5
IXI: L+h$_{16}$ − AD	−0.49	[−2.02, +1.04]	−2.40 to +0.96	0.42	1.00	1/5
OASIS: L+h$_{16}$ − constant	−10.76	[−11.80, −9.72]	−11.63 to −9.39	9×10^{-6}	7×10^{-5}	0/5
OASIS: L+h$_{16}$ − FT	−0.05	[−1.95, +1.85]	−1.97 to +1.89	0.95	1.00	2/5
OASIS: L+h$_{16}$ − RI	−1.15	[−2.78, +0.47]	−2.73 to +0.64	0.12	0.48	1/5
OASIS: L+h$_{16}$ − AD	+2.26	[+0.17, +4.35]	−0.28 to +3.88	0.04	0.20	4/5
Classification, ΔAUC						
IXI sex: L+h$_{16}$ − FT	+0.065	[−0.045, +0.174]	−0.027 to +0.163	0.18	0.78	4/5
IXI sex: L+h$_{16}$ − SL	+0.176	[+0.125, +0.227]	+0.137 to +0.243	7×10^{-4}	6×10^{-3}	5/5
IXI sex: L+h$_{16}$ − RI	+0.031	[−0.025, +0.087]	−0.029 to +0.077	0.20	0.78	4/5
IXI sex: L+h$_{16}$ − AD	−0.025	[−0.050, −0.000]	−0.040 to +0.010	0.05	0.30	1/5
OASIS sex: L+h$_{16}$ − FT	+0.064	[−0.071, +0.200]	−0.065 to +0.224	0.26	0.78	4/5
OASIS sex: L+h$_{16}$ − SL	+0.189	[+0.117, +0.261]	+0.156 to +0.293	2×10^{-3}	0.02	5/5
OASIS sex: L+h$_{16}$ − RI	+0.041	[−0.024, +0.106]	−0.009 to +0.130	0.16	0.78	4/5
OASIS sex: L+h$_{16}$ − AD	+0.005	[−0.040, +0.050]	−0.043 to +0.050	0.77	0.78	3/5
OASIS CDR: L+h$_{16}$ − FT	−0.042	[−0.074, −0.009]	−0.080 to −0.011	0.02	0.16	0/5

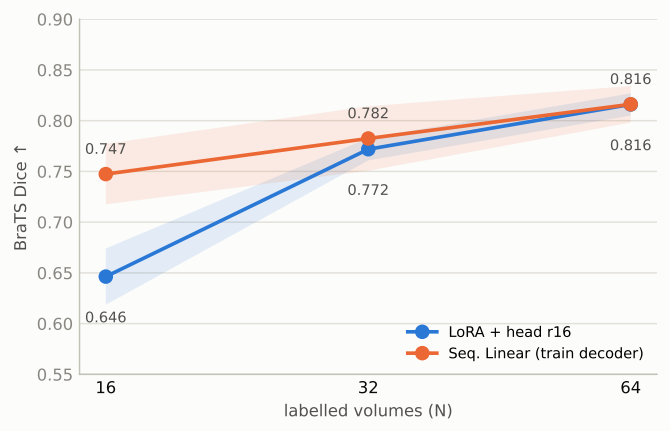


Fig. 2. BraTS Dice against the number of labelled volumes for LoRA + head (r=16) and Sequential Linear (decoder trained on a frozen encoder); mean ± SD over five seeds. Training the decoder leads by 0.10 Dice at 16 shots and the gap is gone at 64.

the strategies that train nothing in the encoder and fit the 513-parameter head by gradient descent return a constant, while 56K low-rank parameters in the encoder learn the task and do so better than releasing all 14M at the same learning rate. What the result does not show, and what we first read into it, is that frozen features cannot express brain age; the closed-form probe of Sec. V-F shows that they largely can.

E. Both Patterns Replicate on Held-Out Task Families

The two findings above each rested on a single task and a single cohort. Table VIII and Fig. 5 add two segmentation tasks and four global tasks chosen to share as little as possible with the originals (a different pathology, a different site and scanner, a second cohort for both global tasks, and a clinical label), and both patterns hold.

Segmentation. On hippocampus segmentation (one channel, healthy anatomy, a foreground two orders of magnitude larger than a lesion, and volumes that fit inside the inference

TABLE V
LORA RANK ABLATION (LORA + HEAD) AND THE BITFIT COMPARATOR, MEAN ± SD OVER SEEDS 42–46. SEQUENTIAL LINEAR SHOWN FOR REFERENCE.

Config	Dice↑	HD95↓	MAE (yr)	Trainable (T1)
BitFit	0.436 ± 0.031	77.1 ± 9.3	14.66	6,434
$r = 4$	0.754 ± 0.032	49.0 ± 6.6	14.32	23,502
$r = 8$	0.750 ± 0.027	48.8 ± 4.6	14.43	46,938
$r = 16$	$\mathbf{0.772 \pm 0.011}$	$\mathbf{44.8 \pm 2.4}$	14.67	93,810
Seq. Linear	0.782 ± 0.032	43.5 ± 7.1	14.71	8,446,306

TABLE VI
SHOT ABLATION, BRATS DICE, MEAN ± SD OVER SEEDS 42–46.

Method	$N_k = 16$	$N_k = 32$	$N_k = 64$
LoRA + head, r=16	0.646 ± 0.028	0.772 ± 0.011	0.816 ± 0.011
Sequential Linear	0.747 ± 0.030	0.782 ± 0.032	0.816 ± 0.018

window), linear probing leads LoRA by 0.011 Dice on every seed (per-seed range +0.007 to +0.015, $p_{\text{Holm}} = 0.02$), the one segmentation contrast between the two that survives correction. On white-matter hyperintensities, whose test set spans five scanners, the difference is +0.015 ($p = 0.58$); on stroke lesions, now with every strategy trained on ISLES alone, it is −0.050 in LoRA's favour ($p = 0.24$); on BraTS −0.011 ($p = 0.57$). Across the four segmentation tasks the frozen-encoder decoder is therefore never worse than LoRA and better on one, over tasks that differ in modality count, pathology, foreground fraction, input scope and scanner (Table III). Full fine-tuning at the shared learning rate leads LoRA on hippocampus (+0.025) and WMH (+0.108), both on every seed and both surviving correction, and is level with it on BraTS and ISLES.

Figure 4 shows representative cases: on each task the two frozen-encoder strategies produce masks of similar quality, differing in detail rather than in kind, which is what a 0.011-Dice margin looks like.

Global tasks. On IXI sex classification, LoRA reaches AUC 0.878 ± 0.079, against 0.814 for full fine-tuning at the shared learning rate (ΔAUC +0.065, $p = 0.18$). The gradient-trained linear probe and BitFit again produce a *constant* decision (balanced accuracy 0.500 on every seed for the probe and on four of five for BitFit) with AUCs of 0.70 and 0.74, well above chance: the frozen features rank subjects by sex, but gradient descent on a 1K-parameter head over raw pooled features never finds a usable boundary. LoRA leads that probe by 0.176 AUC on every seed ($p_{\text{Holm}} = 0.006$). Sec. V-F shows that a closed-form fit of the same head reaches LoRA's AUC exactly.

A second cohort. OASIS-1 repeats both global tasks on a different scanner, a single T1 channel and a wider age range. LoRA reaches brain-age MAE 11.95 ± 0.68 yr ($R^2 = 0.60$) against a 22.7 yr training-mean constant baseline (every seed, $p_{\text{Holm}} = 7\times10^{-5}$) and sex AUC 0.920 ± 0.035; the gradient probe and BitFit return constants on brain age (MAE 23.0) and on four of five seeds for sex (balanced accuracy 0.50). Full fine-tuning matches LoRA on OASIS brain age (Δ −0.05 yr,

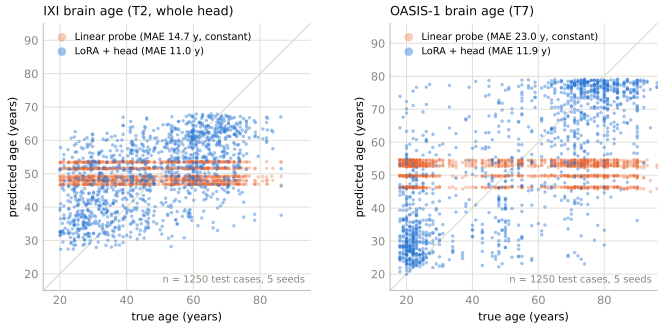


Fig. 3. Predicted against true brain age on the held-out test sets, all five seeds pooled (1250 cases per cohort), LoRA + head (r=16) against the linear probe on the same frozen encoder. Whole-head 96^3 input. The probe's output barely varies with the input (its spread is seed-to-seed, not case-to-case); LoRA's tracks age on IXI and OASIS-1 alike, with the compression towards the training mean expected at 32 labelled subjects.

TABLE VII
IXI BRAIN AGE WITH WHOLE-HEAD INPUT (96^3) VERSUS 64^3 PATCHES, MEAN ± SD OVER SEEDS 42–46, 250 TEST SUBJECTS PER SEED. PRED. SD IS THE STANDARD DEVIATION OF THE PREDICTED AGES; THE TRUE SD IS 16.5 YR, SO A VALUE NEAR ZERO DENOTES A CONSTANT PREDICTOR.

	whole head (96^3)			patch
Method	MAE (yr)↓	R^2 ↑	pred. SD	MAE
LoRA + head, r=16	$\mathbf{10.99 \pm 1.27}$	$\mathbf{+0.31 \pm 0.15}$	10.56	14.67
Sequential FT	12.67 ± 0.69	$+0.13 \pm 0.05$	9.39	14.65
Sequential Linear	14.68 ± 0.22	-0.04 ± 0.04	0.13	14.71
BitFit	14.67 ± 0.24	-0.04 ± 0.04	0.11	14.66
constant (train mean)	14.67	0.00	0	14.67

$p = 0.95$) and sex (+0.064 AUC, $p = 0.26$). Dementia classification (CDR>0, T9) is learned by none of the strategies at 32 labelled subjects (AUC 0.42–0.56, balanced accuracy 0.49–0.52), which bounds what the protocol can detect and is reported as such.

A second adapter family. Houlsby-style bottleneck adapters, the other standard way of moving a frozen encoder, behave like LoRA on the global tasks (Adapter column of Table VIII; paired contrasts in Tables III and IV): IXI brain age 11.48 ± 2.15 yr ($p = 0.42$ against LoRA), OASIS brain age 9.68 ± 1.08 yr, 2.3 yr better than LoRA on four of five seeds ($p = 0.04$, $p_{\text{Holm}} = 0.20$), and sex AUC 0.903 and 0.915 ($p_{\text{Holm}} = 0.30$ and 0.78). On segmentation the two are level on hippocampus and WMH and the adapters trail on the two small-lesion tasks, by 0.017 Dice on BraTS ($p = 0.42$) and 0.104 on stroke lesions ($p = 0.03$, $p_{\text{Holm}} = 0.49$); with five seeds only the direction, consistent on every seed for ISLES, can be claimed. We note, without testing it, that a serial bottleneck rewrites every feature map it is inserted into whereas a parallel low-rank branch adds to it, which could matter where the signal is a few voxels wide.

F. Head Size or Task Type? Breaking the Confound

The nine tasks confound head size with output type: every large-head task is segmentation and every small-head task is a global prediction, and the strategies share one learning rate.

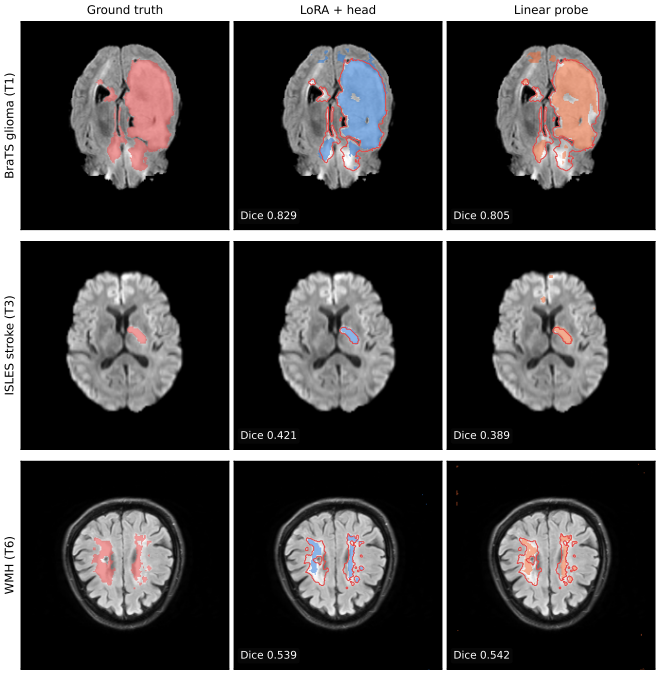


Fig. 4. Qualitative segmentation on held-out test cases (seed-42 split), one per task, chosen as the case whose LoRA Dice is closest to that task's median rather than its best. Left: ground truth (red). Middle and right: LoRA + head (r=16) and the linear probe (decoder trained on the frozen encoder), with the ground-truth outline and the whole-volume per-case Dice. Slice through the largest ground-truth cross-section.

Three sets of control runs separate these (Table IX, paired contrasts in Table X).

A small head on a segmentation task. Keeping the encoder *and* decoder frozen and training only the 66-parameter output projection collapses on all four segmentation tasks (Dice 0.23, 0.03, 0.11 and 0.01; every seed, $p_{\text{Holm}} \leq 0.008$). The large-head half of the rule is therefore about the decoder: segmentation needs it trained, and a frozen encoder beneath it is enough.

A larger head, and a better-fitted small one, on the global tasks. Replacing the 513-parameter head by a two-layer MLP (131K parameters) and training it by gradient descent does not help: it stays constant on brain age and is 0.2–0.3 AUC *below* LoRA on sex. Fitting the original linear head in closed form instead, ridge regression or L2-regularised logistic regression on z-scored pooled features with the penalty chosen on validation, changes the picture. On IXI the closed-form probe reaches sex AUC 0.878 ± 0.024, identical to LoRA's mean ($\Delta = 0.000$, $p = 0.99$), and brain-age MAE 11.70 ± 0.63 yr, 0.7 yr behind LoRA on every seed but not significantly ($p = 0.13$). On OASIS it reaches 15.46 yr and AUC 0.845, and here LoRA's lead is real: 3.5 yr on every seed ($p_{\text{Holm}} = 0.03$) and 0.075 AUC ($p = 0.02$, $p_{\text{Holm}} = 0.13$). The same probe fitted on all 249 IXI subjects reaches 9.1 yr where the gradient-trained head stays at 14.6 (Sec. V-G). The constant output of Sec. V-D is thus an optimisation failure: a tiny head trained by AdamW at 10^{-3} on unstandardised features whose between-subject spread is about 1% of their magnitude never leaves the training mean. Partial fine-tuning, which releases the deepest encoder

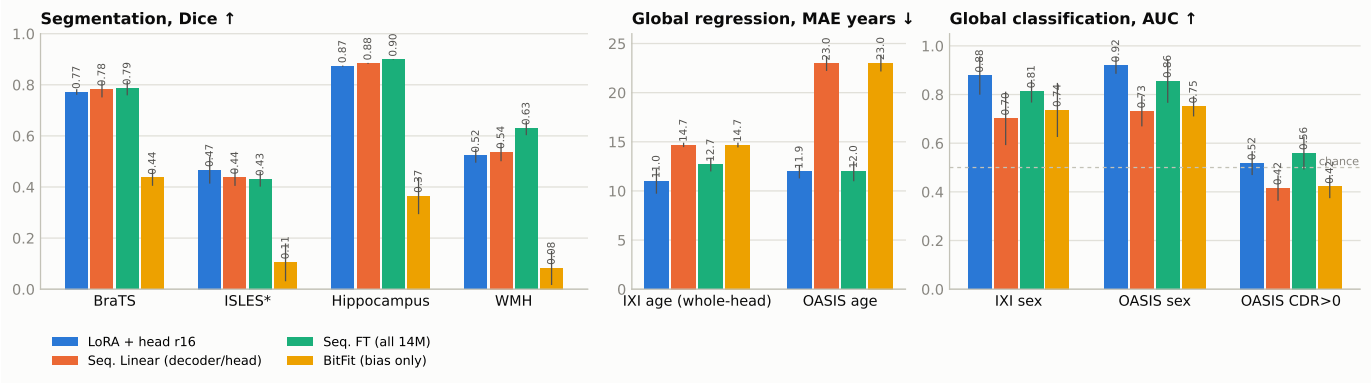


Fig. 5. Four strategies on all nine tasks at $N_k = 32$, mean ± SD over five seeds, at the shared learning rate. Left: segmentation tasks (Dice), where the frozen-encoder decoder is level with LoRA or ahead. Middle and right: global tasks (MAE in years; AUC), where the gradient-trained linear probe and BitFit return constant outputs; the closed-form probe of Table IX does not. Dashed line: chance.

TABLE VIII

ALL NINE TASKS, $N_k = 32$, MEAN ± SD OVER SEEDS 42–46. SEGMENTATION REPORTS DICE↑; BRAIN AGE MAE IN YEARS↓; CLASSIFICATION AUC↑. "CONST." MARKS A METHOD WHOSE OUTPUT DOES NOT VARY WITH THE INPUT (PREDICTION SD < 0.5 YR FOR BRAIN AGE; BALANCED ACCURACY WITHIN 0.02 OF 0.500 FOR CLASSIFICATION). RI = LORA + HEAD ON A RANDOMLY INITIALISED BACKBONE (SEC. V-K). ALL ISLES ENTRIES ARE SINGLE-TASK RUNS. ADAPTER = HOULSBY-STYLE BOTTLENECK ADAPTERS AT LORA'S BUDGET (SEC. IV-D). TRAINABLE COUNTS FOR THE GLOBAL TASKS ARE FOR CLASSIFICATION AND INCLUDE ITS 1,026-PARAMETER HEAD. BOLD: BEST OF THE FIVE STRATEGIES.

Task	Type	LoRA + head, r=16	Sequential Linear	Sequential FT	BitFit	
Tasks with a large task-specific decoder (Dice↑)						
T1 BraTS glioma	seg, 3 ch	0.772 ± 0.011	0.782 ± 0.032	$\mathbf{0.787 \pm 0.028}$	0.436 ± 0.031	0
T3 ISLES stroke	seg, 3 ch	$\mathbf{0.466 \pm 0.053}$	0.417 ± 0.032	0.409 ± 0.057	0.105 ± 0.074	0
T4 MSD hippocampus	seg, 1 ch	0.873 ± 0.002	0.884 ± 0.002	$\mathbf{0.898 \pm 0.001}$	0.365 ± 0.071	0
T6 WMH 2017	seg, 2 ch, 5 scanners	0.522 ± 0.027	0.537 ± 0.036	$\mathbf{0.630 \pm 0.027}$	0.083 ± 0.067	0
Tasks with only a ∼1K-parameter head						
T2 IXI brain age	reg, MAE yr↓	$\mathbf{10.99 \pm 1.27}$	14.68 (const.)	12.67 ± 0.69	14.67 (const.)	1
T7 OASIS brain age	reg, MAE yr↓	11.95 ± 0.68	22.96 (const.)	12.00 ± 1.01	23.01 (const.)	**9**
T5 IXI sex	cls, AUC↑	0.878 ± 0.079	0.702 (const.)	0.814 ± 0.047 (const. 1/5)	0.737 ± 0.111 (const. 4/5)	**0.903** ±
T8 OASIS sex	cls, AUC↑	$\mathbf{0.920 \pm 0.035}$ (const. 1/5)	0.731 ± 0.062 (const. 4/5)	0.856 ± 0.090	0.752 (const.)	0
T9 OASIS CDR>0	cls, AUC↑	$\mathbf{0.518 \pm 0.049}$ (const. 2/5)	0.416 (const.)	0.560 ± 0.068 (const. 3/5)	0.421 (const.)	0.456 ±
Trainable (seg / global)		93.8K / 56.4K	8.45M / 1.0K	22.6M / 14.1M	6.4K / 5.0K	9

stage together with everything downstream, lands between the probe and LoRA on brain age and OASIS sex, slightly above both on IXI sex (+0.029 AUC, $p = 0.25$), and level with LoRA on BraTS; none of these differences is significant.

A learning rate per strategy. The shared rate of 10^{-3} was chosen for LoRA. Table XI re-runs the five main strategies at 10^{-4}, 3×10^{-4} and 10^{-3} on BraTS, IXI brain age and IXI sex (seeds 42–44) and picks, per strategy and task, the rate with the best validation metric. BitFit and the adapters keep 10^{-3} on every task, and LoRA keeps it on BraTS and IXI brain age and drops to 3×10^{-4} on IXI sex, where the change costs 0.013 AUC on the test set. Full fine-tuning prefers 10^{-4} on BraTS and IXI sex and improves substantially: Dice $0.798 \rightarrow 0.835$, now 0.06 above LoRA, and AUC $0.822 \rightarrow 0.911$, now 0.03 above it; on IXI brain age it chooses 3×10^{-4} and stays 1.9 yr behind LoRA. The shared-rate comparison therefore understates full fine-tuning on two of three tasks. The ordering among the frozen-backbone strategies is unchanged by the sweep, and LoRA's rate sits at the top of the grid on two tasks, so a wider grid could move it too.

What survives the controls is narrower than the rule we first stated. Segmentation needs its decoder trained and nothing more, and no strategy that trains only a small head can do it. The global tasks are learnable from frozen features once the head is fitted properly; moving the encoder is worth about 3.5 yr and 0.08 AUC on the harder cohort and nothing measurable on the easier one. Full fine-tuning, given its own learning rate, is the most accurate single-task strategy on segmentation and sex, and its disadvantage is forgetting (Sec. V-I), not accuracy.

G. How Far Below Full Data?

Table XII trains the same strategies on every labelled subject available outside the seed-42 validation and test sets, on the four tasks that have any. With 908 BraTS volumes LoRA + head reaches Dice 0.859, the linear probe 0.849 and full fine-tuning 0.865, against 0.772, 0.782 and 0.787 at 32 shots; with 249 IXI subjects LoRA's brain-age MAE falls from 10.99 to 8.33 yr (R^2 0.31 to 0.55) and its sex AUC from 0.878 to 0.994; with 106 OASIS subjects from 11.95 to 8.27 yr (R^2 0.60 to 0.79) and 0.920 to 0.971. Two things follow. The 32-shot results are not near saturation: they sit 0.09 Dice and 2.7–3.7 yr below what the same recipe reaches with all the data, which is the regime the paper is about. And the spread

TABLE IX
CONTROL ARMS THAT SEPARATE HEAD SIZE FROM TASK TYPE, $N_k = 32$, MEAN ± SD OVER SEEDS 42–46. "SMALL HEAD": FOR SEGMENTATION THE FROZEN ENCODER AND DECODER WITH ONLY THE 66-PARAMETER OUTPUT PROJECTION TRAINED; FOR THE GLOBAL TASKS THE 513-PARAMETER HEAD FITTED IN CLOSED FORM (RIDGE OR LOGISTIC REGRESSION ON Z-SCORED FROZEN FEATURES, PENALTY CHOSEN ON VALIDATION). "LARGE HEAD": FOR SEGMENTATION THE FULL DECODER TRAINED ON THE FROZEN ENCODER (SEQUENTIAL LINEAR); FOR THE GLOBAL TASKS A TWO-LAYER MLP HEAD (131K PARAMETERS) TRAINED BY GRADIENT DESCENT. "PARTIAL FT": THE DEEPEST ENCODER STAGE AND EVERYTHING DOWNSTREAM. "CONST." AS IN TABLE VIII.

Task	LoRA + head, r=16	Small head only	Large head only	Partial FT
T1 BraTS	0.772 ± 0.011	0.230 ± 0.134	0.782 ± 0.032	0.775 ± 0.037
T3 ISLES	0.466 ± 0.053	0.029 ± 0.034	0.417 ± 0.032	n/a
T4 hippocampus	0.873 ± 0.002	0.114 ± 0.038	0.884 ± 0.002	n/a
T6 WMH	0.522 ± 0.027	0.008 ± 0.009	0.537 ± 0.036	n/a
T2 IXI age	10.99 ± 1.27	11.70 ± 0.63	14.72 (const.)	11.23 ± 1.28
T7 OASIS age	11.95 ± 0.68	15.46 ± 1.02	22.91 (const.)	13.20 ± 0.90
T5 IXI sex	0.878 ± 0.079	0.878 ± 0.024	0.671 (const.)	0.907 ± 0.048
T8 OASIS sex	0.920 ± 0.035 (const. 1/5)	0.845 ± 0.016	0.642 (const.)	0.910 ± 0.027

TABLE X
PAIRED CONTRASTS FOR THE CONTROL ARMS (COLUMNS AS IN TABLE III; DICE FOR SEGMENTATION, Δ|ERROR| IN YEARS FOR BRAIN AGE, ΔAUC FOR SEX). HOLM-ADJUSTED WITHIN THIS TABLE.

Contrast	Δ	95% CI	per-seed range	p	p_{Holm}	seeds
BraTS: head only − L+h$_{16}$	−0.542	[−0.697, −0.387]	−0.733 to −0.404	6×10^{-4}	8×10^{-3}	0/5
ISLES: head only − L+h$_{16}$	−0.437	[−0.539, −0.335]	−0.516 to −0.308	3×10^{-4}	4×10^{-3}	0/5
Hippocampus: head only − L+h$_{16}$	−0.759	[−0.806, −0.712]	−0.808 to −0.715	1×10^{-6}	3×10^{-5}	0/5
WMH: head only − L+h$_{16}$	−0.514	[−0.549, −0.479]	−0.542 to −0.483	2×10^{-6}	4×10^{-5}	0/5
BraTS: partial FT − L+h$_{16}$	+0.003	[−0.049, +0.055]	−0.041 to +0.057	0.89	1.00	3/5
IXI age: ridge probe − constant	−2.97	[−3.62, −2.32]	−3.50 to −2.12	2×10^{-4}	3×10^{-3}	0/5
IXI age: ridge probe − L+h$_{16}$	+0.71	[−0.33, +1.76]	+0.13 to +2.19	0.13	0.79	5/5
IXI age: MLP head − L+h$_{16}$	+3.74	[+2.17, +5.31]	+2.16 to +5.51	3×10^{-3}	0.03	5/5
IXI age: partial FT − L+h$_{16}$	+0.24	[−0.39, +0.87]	−0.40 to +0.77	0.35	1.00	3/5
OASIS age: ridge probe − constant	−7.25	[−8.08, −6.42]	−8.03 to −6.38	2×10^{-5}	3×10^{-4}	0/5
OASIS age: ridge probe − L+h$_{16}$	+3.51	[+2.07, +4.95]	+2.24 to +4.73	2×10^{-3}	0.03	5/5
OASIS age: MLP head − L+h$_{16}$	+10.97	[+9.54, +12.39]	+9.53 to +12.55	3×10^{-5}	5×10^{-4}	5/5
OASIS age: partial FT − L+h$_{16}$	+1.26	[−0.40, +2.92]	−0.47 to +3.26	0.10	0.73	4/5
IXI sex: ridge probe − L+h$_{16}$	−0.000	[−0.096, +0.095]	−0.089 to +0.113	0.99	1.00	2/5
IXI sex: MLP head − L+h$_{16}$	−0.207	[−0.320, −0.093]	−0.367 to −0.133	7×10^{-3}	0.07	0/5
IXI sex: partial FT − L+h$_{16}$	+0.029	[−0.030, +0.088]	−0.026 to +0.093	0.25	1.00	3/5
OASIS sex: ridge probe − L+h$_{16}$	−0.075	[−0.127, −0.023]	−0.106 to −0.002	0.02	0.13	0/5
OASIS sex: MLP head − L+h$_{16}$	−0.278	[−0.451, −0.105]	−0.496 to −0.150	0.01	0.10	0/5
OASIS sex: partial FT − L+h$_{16}$	−0.010	[−0.042, +0.021]	−0.039 to +0.022	0.41	1.00	2/5

TABLE XI
LEARNING RATE CHOSEN PER STRATEGY. SEEDS 42–44 AT 10^{-4}, 3×10^{-4} AND 10^{-3}; FOR EACH STRATEGY AND TASK THE RATE WITH THE BEST MEAN VALIDATION METRIC IS SELECTED AND THE TEST METRIC AT THAT RATE IS GIVEN BESIDE THE TEST METRIC AT THE SHARED RATE 10^{-3}. L = LORA + HEAD (r=16), SL = SEQUENTIAL LINEAR, FT = SEQUENTIAL FT, BF = BITFIT, AD = BOTTLENECK ADAPTERS.

	L		SL		FT		BF		AD	
Task	shared	chosen	shared	chosen	shared	chosen	shared	chosen	shared	chosen
BraTS Dice	0.777	0.777 (1e-3)	0.789	0.797 (3e-4)	0.798	0.835 (1e-4)	0.441	0.441 (1e-3)	0.754	0.754 (1e-3)
IXI age MAE	10.22	10.22 (1e-3)	14.65	14.65 (1e-3)	12.20	12.12 (3e-4)	14.63	14.63 (1e-3)	9.99	9.99 (1e-3)
IXI sex AUC	0.876	0.863 (3e-4)	0.717	0.717 (1e-3)	0.822	0.911 (1e-4)	0.743	0.743 (1e-3)	0.895	0.895 (1e-3)

between strategies stays small: on BraTS the probe is now 0.010 Dice below LoRA and full fine-tuning 0.006 above it, so 93.8K low-rank parameters stay within 0.006 Dice of training all 22.6M even at 908 volumes; on all four global tasks LoRA still matches or beats full fine-tuning (8.33 vs 8.50 yr, 8.27 vs 9.45, AUC 0.994 vs 0.958 and 0.971 vs 0.953). The closed-form probe (Sec. V-F) fitted on the same 249 IXI subjects reaches 9.14 yr and AUC 0.978, against 14.61 and 0.88 for the gradient-trained head on the same frozen features, and 13.2 yr and 0.923 on the 106 OASIS subjects. These are single runs and are reported as reference points, not as a comparison.

H. Does the Rule Depend on the Backbone?

Everything above is measured on one convolutional backbone. Table XIII repeats the four strategies that define the decision rule on a Swin UNETR with a self-supervised transformer encoder [28], [29], on one large-head task and two small-head tasks, with the recipe unchanged and the same three seeds on both backbones.

The large-head half transfers intact. On BraTS the Swin probe is within 0.004 Dice of Swin LoRA (0.816 vs 0.820; $p = 0.71$ over three seeds), full fine-tuning leads by 0.031

TABLE XII
FULL-DATA REFERENCE, SEED 42: THE 32-SHOT RESULT (SEED 42) BESIDE THE SAME STRATEGY TRAINED ON EVERY LABELLED SUBJECT OUTSIDE THAT SEED'S VALIDATION AND TEST SETS (N_{FULL}; ONE RUN). BRATS RUNS 10 EPOCHS OVER 908 VOLUMES; THE GLOBAL TASKS KEEP THE 100-EPOCH RECIPE. ISLES, HIPPOCAMPUS, WMH AND CDR HAVE NO SUBJECTS OUTSIDE THE FEW-SHOT PROTOCOL'S SPLITS.

			LoRA + head, r=16		Sequential Linear		Sequential FT	
Task	Metric	N_{full}	32 shots	full	32 shots	full	32 shots	full
T1 BraTS glioma	Dice↑	908	0.783	0.859	0.795	0.849	0.813	0.865
T2 IXI brain age	MAE yr↓	249	9.15	8.33	14.64	14.61	12.01	8.50
T5 IXI sex	AUC↑	249	0.918	0.994	0.781	0.882	0.892	0.958
T7 OASIS brain age	MAE yr↓	106	12.38	8.27	23.04	22.96	12.86	9.45
T8 OASIS sex	AUC↑	106	0.919	0.971	0.753	0.771	0.863	0.953

on every seed ($p = 0.02$, $p_{\text{Holm}} = 0.13$) and BitFit collapses (0.421; $p_{\text{Holm}} = 0.02$): the U-Net ordering at a higher absolute level. The parameter ratio is also the same, a 54.1M-parameter decoder against 618K low-rank parameters, 88×.

The small-head half transfers for classification and not for regression. On IXI sex, Swin LoRA has the highest AUC (0.871 against 0.644 for the probe and 0.764 for full fine-tuning), though on two of three seeds it assigns one class to every subject at the 0.5 threshold; the probe and full fine-tuning are at or near a constant output on two of three seeds; and BitFit, which on a transformer moves every LayerNorm offset and attention bias, is the one strategy whose decisions vary with the input on every seed (AUC 0.817, balanced accuracy 0.70); with three seeds none of the Swin sex contrasts is significant (Table XIV). On IXI brain age no strategy learns: all four return a constant (MAE 14.6–14.8 yr, prediction SD < 0.4 yr), LoRA and full fine-tuning included. The frozen features show why. The pooled 768-dimensional bottleneck feature of the pretrained Swin encoder is almost the same vector for every subject: mean pairwise cosine 0.99997 over the 32 training heads, a between-subject SD of 0.8% of the feature magnitude (the U-Net's pooled feature: 1.5%). The mean-squared-error recipe tuned on the U-Net, 100 epochs at a learning rate of 10^{-3}, never leaves the constant-prediction solution from that starting point, and LoRA's $+0.16$ yr against the constant baseline ($p = 0.17$) is the wrong sign. We did not retune it, because a per-backbone recipe search would make the comparison between backbones uninterpretable.

What the replication establishes is the segmentation half: on either backbone, training the inherited decoder is as accurate as LoRA. On the global tasks it mainly shows how recipe-dependent the small-head results are, which Sec. V-F then explains; the closed-form probe was not run on the Swin features.

I. Forgetting

Every method with a frozen backbone, adapter-isolated or linear-probed, has BWT exactly zero on every seed, because nothing shared is updated. Every method that updates the backbone forgets. On the T1→T2 sequence unregularised fine-tuning loses 0.331 ± 0.148 Dice on BraTS after learning patch-based brain age, EWC 0.321 ± 0.181, LwF 0.475 ± 0.171 and replay 0.400 ± 0.103; across seeds none of the three differs from plain fine-tuning (EWC $p = 0.94$, LwF 0.13, replay 0.46; Table XVI), so the claim that survives is that none prevents forgetting, not that any makes it worse. This sequence has a weakness as a test of the remedies: its second task is patch-based brain age, which no strategy learns (Sec. V-D), so the backbone is being pulled towards a task with no signal.

Table XV therefore repeats the two-task sequence with a second task that is learnable, whole-head IXI brain age at 96^3, starting every run from the same five BraTS checkpoints of sequential fine-tuning, and adds the output-distillation form of LwF [5] and labelled rehearsal to the feature-based forms used above. Fine-tuning now loses 0.542 ± 0.203 Dice. EWC reduces the loss at every penalty, and at $\lambda = 10^2$ reliably: BWT -0.377 against -0.542, better on every seed ($p = 0.006$, $p_{\text{Holm}} = 0.04$, the only remedy that survives correction), for a 0.8 yr cost on the new task; at 10^4 the loss is 0.294 ($p = 0.02$, $p_{\text{Holm}} = 0.09$) and the new task costs 1.1 yr. The four distillation and rehearsal variants land between -0.34 and -0.49, none distinguishable from fine-tuning. No remedy brings BWT near zero, and every one still loses more than a third of the first task's Dice.

The three-task sequence sharpens this (Table XVII, Fig. 6). After learning ISLES, full fine-tuning's BraTS Dice falls from 0.779 ± 0.028 to 0.202 ± 0.172 (and 0.247 ± 0.280 after brain age), with the loss ranging from near zero to almost everything across seeds. A second segmentation task damages the first more than the heterogeneous regression task did, and how much is unpredictable. ISLES itself then loses 0.18 after brain age. Reversing the sequence (T2→T3→T1) shows the effect is symmetric: ISLES, now learned second, falls from 0.401 ± 0.085 to 0.010 ± 0.005 once BraTS is learned after it. Whichever segmentation task comes first, the next one erases it. The reversed two-task sequence (T2→T1) is uninformative on this point, because patch-based brain age sits at the constant baseline before anything can be forgotten (MAE 14.5–14.7 before, 14.6–14.8 after). The frozen methods score identically before and after every step, and ISLES Dice at learn is comparable across them (0.440 LoRA, 0.437 linear, 0.431 full fine-tuning).

J. What Each Strategy Costs to Deploy

Accuracy is one axis; a service that accumulates tasks also pays in storage, training time and hardware. Table XVIII gives the measured figures for the segmentation configuration, which is the largest of our tasks.

Storage is where the strategies separate. A LoRA adapter with its output head occupies 366 KB on disk, so a site running

TABLE XIII
SECOND BACKBONE: THE FOUR STRATEGIES THAT DEFINE THE DECISION RULE ON THE FOMO 3D U-NET AND ON A SWIN UNETR WITH A SELF-SUPERVISED TRANSFORMER ENCODER, $N_k = 32$, MEAN ± SD OVER SEEDS 42–44 ON BOTH BACKBONES (SO THE U-NET COLUMNS DIFFER SLIGHTLY FROM TABLE VIII). "CONST." AS IN TABLE VIII. TRAINABLE PARAMETERS, U-NET / SWIN, SEGMENTATION AND WHOLE HEAD (WHOLE-HEAD COUNTS ARE FOR BRAIN AGE AND INCLUDE THE POOLED HEAD): LORA 93.8K / 618K AND 55.8K / 485K; SEQUENTIAL LINEAR 8.4M / 54.1M AND 0.5K / 0.8K; SEQUENTIAL FT 22.6M / 62.2M AND 14.1M / 8.1M; BITFIT 6.4K / 21.7K AND 4.5K / 22.4K. BOLD: BEST PER COLUMN.

	T1 BraTS, Dice↑		T2 IXI brain age, MAE yr↓		T5 IXI sex, AUC↑	
Strategy	U-Net	Swin UNETR	U-Net	Swin UNETR	U-Net	Swin UNETR
LoRA + head (r=16)	0.777 ± 0.005	0.820 ± 0.022	$\mathbf{10.22 \pm 1.00}$	14.72 (const.)	$\mathbf{0.876 \pm 0.097}$	0.871 ± 0.044 (const. 2/3)
Sequential Linear	0.789 ± 0.006	0.816 ± 0.030	14.65 (const.)	14.64 (const.)	0.717 (const.)	0.644 ± 0.061 (const. 2/3)
Sequential FT	$\mathbf{0.798 \pm 0.031}$	$\mathbf{0.851 \pm 0.014}$	12.20 ± 0.21	14.76 (const.)	0.822 ± 0.061 (const. 1/3)	0.764 (const.)
BitFit	0.441 ± 0.042	0.421 ± 0.031	14.63 (const.)	14.69 (const.)	0.743 (const.)	$\mathbf{0.817 \pm 0.071}$

TABLE XIV
PAIRED CONTRASTS ON THE SWIN UNETR, SEEDS 42–44 (COLUMNS AS IN TABLE III; DICE, Δ|ERROR| IN YEARS, ΔAUC).

Contrast	Δ	95% CI	range	p	p_{Holm}	seeds
BraTS (Swin): L+h$_{16}$ − SL	+0.004	[−0.033, +0.040]	−0.008 to +0.020	0.71	0.80	1/3
BraTS (Swin): FT − L+h$_{16}$	+0.031	[+0.011, +0.050]	+0.022 to +0.037	0.02	0.13	3/3
BraTS (Swin): L+h$_{16}$ − BF	+0.399	[+0.314, +0.484]	+0.363 to +0.431	2×10^{-3}	0.02	3/3
IXI age (Swin): L+h$_{16}$ − constant	+0.16	[−0.17, +0.50]	+0.06 to +0.32	0.17	0.69	3/3
IXI sex (Swin): L+h$_{16}$ − SL	+0.227	[−0.033, +0.487]	+0.164 to +0.348	0.06	0.32	3/3
IXI sex (Swin): L+h$_{16}$ − FT	+0.107	[−0.327, +0.541]	−0.044 to +0.299	0.40	0.80	2/3
IXI sex (Swin): L+h$_{16}$ − BF	+0.054	[−0.089, +0.197]	−0.005 to +0.110	0.25	0.74	2/3

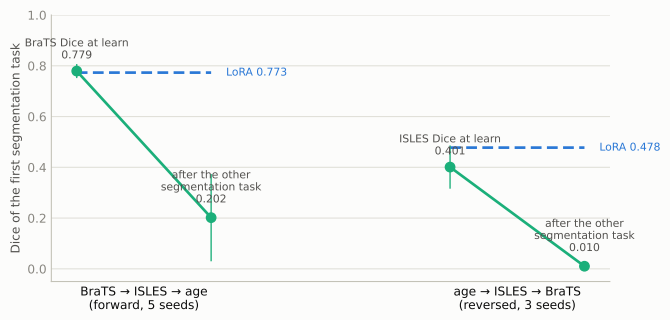


Fig. 6. Under full fine-tuning, the first segmentation task in a three-task sequence is erased by the second whichever order the tasks arrive in: BraTS Dice 0.78→0.20 after ISLES, 0.25 after brain age (forward, five seeds); ISLES Dice 0.40→0.01 after BraTS (reversed, three seeds). Dashed: LoRA + head on the same task, which by construction is unchanged by later tasks.

five tasks stores one 87 MB backbone plus 1.8 MB of adapters (89 MB in all). Linear probing must keep a separate 32 MB decoder per task (161 MB of decoders for five, 248 MB with the backbone), and full fine-tuning a separate 86 MB network per task (431 MB), since nothing is shared once every weight has moved. The ratio grows linearly with the number of tasks, which is the regime a long-lived service is in.

Training time runs the other way. The two strategies that update the fewest parameters are the *slowest* to train: BitFit 48.1 min and LoRA 41.5 min, against 15.8 for linear probing and 15.9 for full fine-tuning. Gradients must still be propagated through the entire network to reach parameters scattered across it, and LoRA's parallel low-rank branches add forward cost; only the optimiser state shrinks. Where linear probing back-propagates no further than the decoder, LoRA and BitFit must traverse the encoder as well. Parameter efficiency here buys storage and task isolation, not training speed. All four fit on one consumer GPU, peaking between 0.9 and 1.7 GB, so hardware is not the discriminator at this scale.

K. Does the Pretrained Checkpoint Help?

Two questions are easily conflated here: whether the checkpoint loads, and whether it helps. On the first, the FOMO checkpoint supplies 59 of the 82 tensors of the segmentation model. Seven have the wrong shape and are left at random initialisation (the input convolution, whose channel count depends on the task; the first convolution of each of the four decoder stages; and the output projection), and the second convolution of each decoder stage is absent from the checkpoint altogether. The encoder proper loads 39 of its 40 tensors. "Pretrained" in this paper therefore means a pretrained encoder beneath a largely re-initialised decoder, which is also what the FOMO baseline code does.

On the second, we repeated the LoRA + head (r=16) configuration on every task with the backbone randomly initialised instead of loaded, changing nothing else (Fig. 7; RI column of Table VIII; paired contrasts in Tables III and IV). On glioma segmentation the two are identical (+0.000 Dice, $p = 0.99$), as on IXI brain age (−0.00 yr, $p = 1.00$), ISLES (−0.030, $p = 0.35$) and hippocampus (+0.002, $p = 0.06$). The checkpoint helps on WMH (+0.024 Dice on four of five seeds, $p = 0.27$), OASIS brain age (1.2 yr, $p = 0.12$) and both sex tasks (+0.03 and +0.04 AUC, $p = 0.20$ and 0.16): consistent in direction, none significant over five seeds. The same holds when the probe or full fine-tuning is initialised either way on BraTS (+0.018 and +0.002 Dice, Table XX).

The closed-form probe gives a cleaner answer about the encoder's features, because it has no optimiser to confound them. Fitted on frozen pretrained features it beats the same fit on frozen random features on every global task and every seed: by 1.5 yr on IXI brain age, 3.0 yr on OASIS brain age, and 0.21 and 0.08 AUC on IXI and OASIS sex (all $p_{\text{Holm}} \leq 0.03$). The pretrained encoder does carry signal that random features lack; once LoRA is allowed to move the encoder, that head start is mostly recovered from random initialisation within 100 epochs.

What is settled is that transfer from a checkpoint cannot be assumed: its loading should be checked tensor by tensor, and a randomly initialised control is cheap enough to run every time. What our results do not measure is the value of the FOMO corpus itself, since they run one checkpoint, one shot count and one adaptation recipe.

TABLE XV
TWO-TASK SEQUENCE WITH A LEARNABLE SECOND TASK: BRATS GLIOMA SEGMENTATION (64^3 PATCHES) FOLLOWED BY WHOLE-HEAD IXI BRAIN AGE (96^3), EVERY RUN INITIALISED FROM THE SAME SEED'S BRATS CHECKPOINT OF SEQUENTIAL FINE-TUNING (SO THE "AT LEARN" COLUMN IS SHARED), $N_k = 32$, MEAN ± SD OVER SEEDS 42–46. EWC: DIAGONAL-FISHER PENALTY AT THREE STRENGTHS. LWF: DISTILLATION OF THE PREVIOUS MODEL'S ENCODER FEATURES (AS IN TABLE II) OR OF ITS BRATS OUTPUT THROUGH THE FROZEN OLD HEAD. REPLAY: A 100-VOLUME BRATS BUFFER USED FOR FEATURE DISTILLATION OR FOR LABELLED REHEARSAL OF THE SEGMENTATION LOSS.

Method	BraTS Dice at learn	Age MAE (yr)	BraTS Dice after age	BWT
Sequential FT	0.787 ± 0.028	12.05 ± 1.02	0.245 ± 0.179	-0.542 ± 0.203
EWC, λ=10^2	0.787 ± 0.028	12.86 ± 1.00	0.409 ± 0.147	-0.377 ± 0.165
EWC, λ=10^3	0.787 ± 0.028	12.76 ± 0.74	0.448 ± 0.217	-0.339 ± 0.214
EWC, λ=10^4	0.787 ± 0.028	13.11 ± 0.66	0.493 ± 0.203	-0.294 ± 0.218
LwF (feature distillation)	0.787 ± 0.028	12.59 ± 0.62	0.300 ± 0.136	-0.487 ± 0.152
LwF (output distillation)	0.787 ± 0.028	12.96 ± 1.01	0.446 ± 0.149	-0.341 ± 0.139
Replay (feature distillation)	0.787 ± 0.028	12.64 ± 0.75	0.371 ± 0.104	-0.416 ± 0.090
Replay (labelled rehearsal)	0.787 ± 0.028	12.59 ± 0.35	0.349 ± 0.180	-0.438 ± 0.195

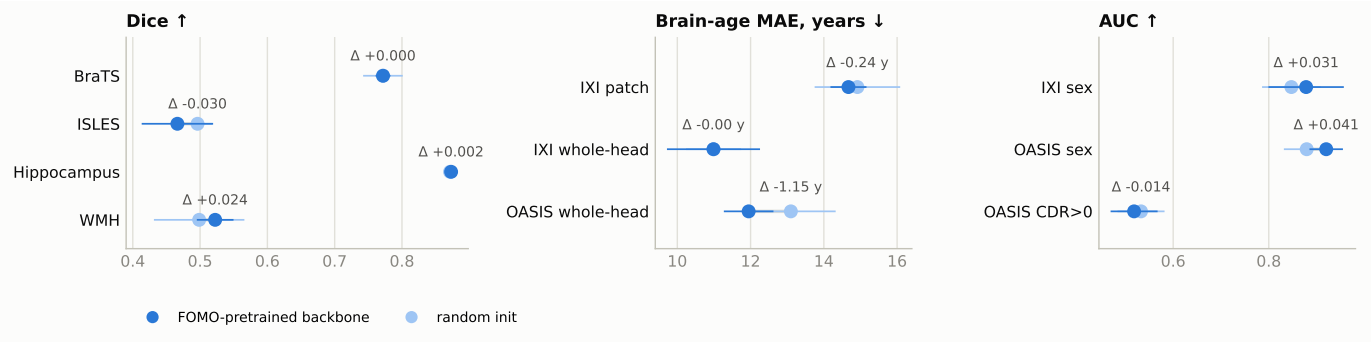


Fig. 7. Pretrained (FOMO checkpoint) versus randomly initialised backbone under the same LoRA + head (r=16) configuration on all nine tasks, mean ± SD over five seeds; Δ is pretrained minus random. Seed-level paired tests are in Tables III and IV.

TABLE XVI
PAIRED BWT CONTRASTS AGAINST SEQUENTIAL FINE-TUNING, SEED AS THE UNIT (COLUMNS AS IN TABLE III). TOP: THE T1→T2 SEQUENCE OF TABLE II. BOTTOM: THE BRATS→WHOLE-HEAD-AGE SEQUENCE OF TABLE XV. POSITIVE FAVOURS THE REMEDY.

Contrast	ΔBWT	95% CI	range	p	p_{Holm}	seeds
BraTS BWT: EWC − FT	+0.010	[−0.347, +0.367]	−0.363 to +0.369	0.94	0.94	2/5
BraTS BWT: LwF − FT	−0.143	[−0.349, +0.062]	−0.354 to +0.057	0.13	0.38	1/5
BraTS BWT: replay − FT	−0.069	[−0.303, +0.165]	−0.318 to +0.126	0.46	0.92	2/5
EWC (λ=10^2) − FT	+0.164	[+0.078, +0.250]	+0.058 to +0.227	6×10^{-3}	0.04	5/5
EWC (λ=10^3) − FT	+0.203	[−0.101, +0.506]	−0.155 to +0.419	0.14	0.69	4/5
EWC (λ=10^4) − FT	+0.248	[+0.079, +0.417]	+0.144 to +0.484	0.02	0.09	5/5
LwF, features − FT	+0.055	[−0.092, +0.201]	−0.088 to +0.175	0.36	0.86	3/5
LwF, outputs − FT	+0.201	[−0.145, +0.546]	−0.128 to +0.479	0.18	0.73	3/5
Replay, features − FT	+0.126	[−0.163, +0.415]	−0.168 to +0.359	0.29	0.86	3/5
Replay, labels − FT	+0.104	[−0.131, +0.338]	−0.176 to +0.324	0.29	0.86	4/5

VI. DISCUSSION

A. Choosing a Strategy

Table XIX states the recommendations compactly; this section gives the reasoning behind them.

The two task families point in opposite directions, each replicated on a task chosen to share little with the original.

When the task head is large, train it; adapters buy economy. Segmentation inherits an 8.45M-parameter decoder that is already task-shaped. Training it on a frozen encoder is a strong and simple baseline that LoRA + head never beats and loses to on hippocampus, and that a frozen decoder with a trainable output projection cannot replace (Dice 0.01–0.23). That the margin is stable across four tasks sharing almost nothing (three modalities, two or one; pathology or healthy anatomy; a lesion occupying <0.1% of voxels or a structure occupying 4.7%; one scanner or five) points to the decoder, not the imaging problem, as what matters. Here the argument for adapters is cost, not accuracy: 93.8K parameters and under 400 KB per task on disk against a 32 MB decoder per task. Where a model will only ever serve one task, full fine-tuning at its own learning rate is more accurate than either, by 0.06 Dice on BraTS and 0.11 on WMH. Not every adapter family qualifies: bottleneck adapters at the same budget match LoRA where the foreground is large and trail on the two small-lesion tasks.

When there is no task head to speak of, fit the head properly before moving the encoder. Brain age and sex both attach a ∼1K-parameter head to pooled encoder features. Trained by gradient descent at the shared recipe, that head returns a constant on both cohorts, and so does bias-only tuning, which is what first led us to conclude that the encoder had to move. The closed-form control shows otherwise: on standardised frozen features the same head matches LoRA on IXI sex and comes within 0.7 yr of it on IXI brain age, and with all 249 subjects it reaches 9.1 yr where the gradient-trained head is still constant. The frozen features carry the information; a tiny head over inputs whose between-subject spread is about 1% of their magnitude is simply a badly conditioned problem for a first-order optimiser at 10^{-3}. What moving the encoder adds is cohort-dependent: 3.5 yr and 0.08 AUC on OASIS, where the scanner, channel count and age range all differ from IXI, and nothing measurable on IXI.

TABLE XVII
THREE-TASK SEQUENCES AT $N_k = 32$. TOP: T1→T3→T2 (BRATS → ISLES → IXI BRAIN AGE, 64^3 PATCHES), MEAN ± SD OVER SEEDS 42–46. BOTTOM: THE REVERSED ORDER T2→T3→T1, SEEDS 42–44. EACH TASK IS SCORED ON ITS OWN HELD-OUT TEST SET WHEN LEARNED AND AGAIN AFTER EACH LATER TASK. ISLES TEST SET: 155 SUBJECTS.

Method	At learn BraTS Dice	At learn ISLES Dice	At learn Age MAE	BraTS Dice after… ISLES	BraTS Dice after… age	ISLES Dice after age
LoRA + head, r=16	0.773 ± 0.011	0.440 ± 0.033	14.51 ± 0.48	=	=	=
Sequential Linear	0.782 ± 0.032	0.437 ± 0.033	14.75 ± 0.26	0.782 ± 0.032	0.782 ± 0.032	0.437 ± 0.033
Sequential FT	0.779 ± 0.028	0.431 ± 0.029	14.69 ± 0.22	0.202 ± 0.172	0.247 ± 0.280	0.254 ± 0.073

Method	At learn (reversed) Age MAE	At learn (reversed) ISLES Dice	At learn (reversed) BraTS Dice	Age MAE after… ISLES	Age MAE after… BraTS	ISLES Dice after BraTS
LoRA + head, r=16	14.57 ± 0.44	0.478 ± 0.031	0.793 ± 0.020	=	=	=
Sequential FT	14.56 ± 0.09	0.401 ± 0.085	0.779 ± 0.049	14.39 ± 0.08	14.42 ± 0.18	0.010 ± 0.005

TABLE XVIII
DEPLOYMENT COST ON THE SEGMENTATION CONFIGURATION (STORAGE EXACT; TRAINING TIME THE MEAN OVER SEEDS 42–44 ON AN OTHERWISE IDLE GPU). "PER TASK" IS WHAT MUST BE STORED FOR EACH ADDITIONAL TASK ONCE THE SHARED 87 MB BACKBONE IS ON DISK; FULL FINE-TUNING SHARES NOTHING, SO ITS BACKBONE IS COUNTED PER TASK.

Strategy	Trainable	Per task	5 tasks	Train
LoRA + head, r=16	93.8 K	**366 KB**	**89 MB**	41.5 min
Linear probe (decoder)	8.45 M	32.2 MB	248 MB	**15.8 min**
Full fine-tuning	22.6 M	86.2 MB	431 MB	15.9 min
BitFit	6.4 K	25 KB	87 MB	48.1 min

TABLE XIX
SELECTING A STRATEGY FOR A NEW TASK. THE CONDITION IS KNOWABLE BEFORE ANY TRAINING: IT ASKS WHETHER THE ARCHITECTURE ALREADY CONTAINS A LARGE MODULE SHAPED FOR THE NEW TASK'S OUTPUT.

Situation	Recommendation
New task reuses a large existing head (e.g. a segmentation decoder)	Train that head on a frozen encoder. It is never less accurate than LoRA and is the simplest option; the encoder need not move. Prefer LoRA only if per-task storage matters: it saves 88× the disk and was not distinguishable from the trained decoder on three of four tasks. If the task is the only one the model will ever serve, full fine-tuning at its own learning rate is more accurate still.
New task has only a small head (a pooled regressor or classifier)	First fit the head in closed form on standardised frozen features; a gradient-trained head at a shared learning rate can return a constant even when the features carry the signal. Then move the encoder (LoRA, bottleneck adapters or partial fine-tuning, which perform alike) where the probe falls short, which here was the cohort farther from the pretraining data.
Prior tasks must not degrade	Freeze the shared weights, by either route. EWC reduces forgetting but does not remove it; LwF and replay, in any of the four forms tried, did no better than unregularised fine-tuning; a second segmentation task erased the first in either order.
More than ∼64 annotated cases available	The accuracy gap between frozen strategies closes; choose on storage and operational fit rather than on Dice.
Comparing strategies	Tune the learning rate per strategy and test at the level of the training run; a shared rate favoured the parameter-efficient methods here, and pooling cases across seeds produced significance that five seeds do not support.
Adopting a pretrained checkpoint	Check which tensors load, and run a randomly initialised control. Ours loaded 59 of 82 segmentation tensors, made no measurable difference under LoRA on four tasks and a small one on the rest, while its frozen features beat random ones under a closed-form probe (Sec. V-K).

LoRA, bottleneck adapters and partial fine-tuning deliver that gain alike, so it is a statement about encoder adaptation, not about LoRA's form.

A shared learning rate is a hidden thumb on the scale. The rate of 10^{-3} was tuned for LoRA and kept for everything. Full fine-tuning at 10^{-4} gains 0.04 Dice on BraTS and 0.09 AUC on IXI sex over its shared-rate result, widening a BraTS lead that was not significant to 0.06 Dice and overtaking LoRA on sex; the frozen-backbone strategies keep 10^{-3}. Comparisons across strategies with different trainable-parameter counts should tune the rate per strategy, and ours before this sweep did not. The forgetting results are unaffected, because they compare one strategy with itself across tasks.

Sex classification shows the mechanism most clearly. The gradient-trained probe's AUC is 0.70–0.73 on the two cohorts, so the frozen features rank subjects by sex well above chance even when the decision is constant; the closed-form fit of the same head reaches 0.878 on IXI and 0.845 on OASIS. The bottleneck we first described as an unusable feature geometry was an unusable optimisation: the information was present and a logistic regression finds it. The lesson for practice is that a small head on frozen features should be fitted in closed form, or at least on standardised inputs with a validated learning rate, before the result is read as a limit of the features.

No replay means no retained prior-task data. Every strategy we recommend trains on the new task's cases alone. Replay-based continual learning requires keeping examples from earlier tasks available for the lifetime of the service, which under HIPAA or GDPR is a governance commitment rather than a storage one; in our results it also did not work, in either the feature-distillation or the labelled-rehearsal form. Freezing obtains the same protection with no retention at all.

Forgetting-free is a property of freezing, not of adapters. Adapter isolation and frozen-encoder probing both have BWT zero; the distinction between them is the trainable-parameter count, not the BWT column. The other side is the finding: no remedy achieved it. On a sequence whose second task is

TABLE XX
PRETRAINED VERSUS RANDOMLY INITIALISED BACKBONE UNDER STRATEGIES OTHER THAN LORA, SEED AS THE UNIT (COLUMNS AS IN TABLE III). PROBE AND FULL FT: GRADIENT-TRAINED ON BRATS AND IXI BRAIN AGE. RIDGE: THE CLOSED-FORM PROBE ON FROZEN FEATURES, PRETRAINED MINUS RANDOM (NEGATIVE FAVOURS PRETRAINED FOR MAE, POSITIVE FOR AUC).

Contrast	Δ	95% CI	range	p	p_{Holm}	seeds
BraTS, probe: pretrained − random	+0.018	[−0.032, +0.069]	−0.028 to +0.064	0.37	0.74	3/5
BraTS, full FT: pretrained − random	+0.002	[−0.032, +0.036]	−0.026 to +0.038	0.88	0.88	3/5
IXI age, full FT: pretrained − random	−0.81	[−2.14, +0.52]	−2.27 to +0.57	0.17	0.50	1/5
IXI age, ridge: pretrained − random	−1.49	[−2.04, −0.93]	−1.99 to −0.77	2×10^{-3}	9×10^{-3}	0/5
IXI sex, ridge: pretrained − random	+0.208	[+0.136, +0.279]	+0.172 to +0.309	1×10^{-3}	9×10^{-3}	5/5
OASIS age, ridge: pretrained − random	−3.01	[−4.03, −1.99]	−3.80 to −2.11	1×10^{-3}	9×10^{-3}	0/5
OASIS sex, ridge: pretrained − random	+0.079	[+0.035, +0.123]	+0.023 to +0.111	8×10^{-3}	0.03	5/5

learnable, EWC reduces the loss of the first task from 0.54 to 0.38 Dice at $\lambda = 10^2$, reliably across seeds, at a cost of 0.8 yr on the second task; the distillation and rehearsal variants, feature-, output- and label-based, are indistinguishable from plain fine-tuning; and a second task of the *same* type as the first erased it whichever came first, with full fine-tuning's loss ranging from near zero to almost all of the Dice across seeds. If prior-task performance matters, freezing is the reliable route; EWC is a partial remedy and the others none.

The 66-parameter lesson. Whether the task's output projection is trainable moves Dice by 0.25 on every seed. The effect is large and the error is easy to make: it is what a literal reading of "freeze the backbone, train the adapter" produces, and on its own it is enough to make LoRA look unsuited to segmentation.

B. Limitations

Pretraining helps inconsistently, and this generalises only as a method. The control (Sec. V-K) finds the checkpoint indistinguishable from random initialisation under LoRA on four tasks and modestly, not significantly, better on the rest, while its frozen features beat random ones under a closed-form probe on every global task. Our results therefore support parameter-efficient adaptation of a frozen backbone; they do not measure the value of large-scale pretraining, and the reader should not generalise the specific per-task findings to other checkpoints or scales. What does generalise is the practice: checking which tensors a checkpoint supplies, and a randomly initialised control, together cost one extra training run.

Nine tasks, but six data sources and one imaging domain. Brain age and sex share subjects and scans within each cohort, so within a cohort they are evidence about output type, not about data; T2 and T7 (and T5 and T8) are the independent replications. All nine tasks are brain MRI. Three were repeated on a second architecture (Sec. V-H), which reproduced the rule's shape but learned brain age with no strategy; other anatomy and modalities are untested.

Five seeds, one shot count for most results, three tasks on the second backbone. Means and SDs are over five training runs (three for the reversed sequences, the learning-rate sweep and the Swin UNETR replication, which also keeps the U-Net's recipe without retuning), and the paired tests use the run as the unit, so a difference must hold across seeds to register and effects smaller than the seed-to-seed spread are not resolved; several contrasts are reported as directions rather than findings for that reason. Only the shot ablation spans 16–64 shots, and only on two methods. The learning-rate sweep covers three rates on three tasks, and LoRA's chosen rate sits at the top of that grid on two of them. The closed-form probe was run on the U-Net features only.

Task identity is given at inference. Adapter selection assumes the task is known; task-agnostic routing is not addressed, and it is the obvious next step for a deployed system.

Two sequences, two orders each. T1→T2 and T1→T3→T2 and their reversals are the only orders evaluated; a sequence longer than three tasks, or one containing two global tasks, is not.

Brain age is improved, not solved. MAE 11.0 yr on IXI ($R^2 = 0.31$) and 12.0 yr on OASIS ($R^2 = 0.60$) beat a constant by a wide margin but are far from the 3–4 yr that large supervised cohorts achieve; 32 labelled subjects and a 96^3 resampling are the obvious constraints. Dementia from 32 labelled T1 volumes (T9) is not learnable by any strategy here, which marks the floor of the protocol rather than a property of any method.

The full-data references are single runs on four tasks. They bound the 32-shot results from above and show that the ordering between strategies survives more data, but with one seed each they cannot support paired tests, and five of the nine tasks have no subjects outside the few-shot protocol's splits, so their ceilings are unknown.

Most tasks are single-task evaluations. T4–T9 are run on their own rather than inside a continual sequence; they test the accuracy patterns, not forgetting, which the three-task sequences cover.

Infarct detection as classification excluded. A binary infarct-detection formulation contrasting ISLES with IXI controls was prepared but is not reported, because the positive class carries DWI, FLAIR and ADC channels and the negative class T1 and T2 with zero-filled channels, so the classes differ by acquisition rather than pathology and any classifier score would be uninterpretable. We use ISLES for lesion segmentation instead, where the cohort is internally consistent.

No augmentation. Few-shot results would likely improve with it for all methods; we omit it to keep the method comparison clean.

VII. CONCLUSION

We compared eleven adaptation strategies across nine brain-MRI tasks, six cohorts, three output types and five seeds, under a protocol with subject-grouped held-out splits, whole-volume inference and the training run as the unit of inference, to answer a question a deployed imaging service actually faces: how should a new task be added when only a few dozen annotations exist and prior-task data is unavailable?

The choice depends on what the new task inherits. Tasks that reuse a large existing head, all four segmentation tasks here, are best served by training that head on a frozen encoder: it is never less accurate than LoRA and better on one task, a frozen decoder with only its output projection trainable collapses, and LoRA is preferable only when per-task storage matters, where it saves 88× the disk. Tasks with only a small head, brain age

and sex from whole-head input on two cohorts, are learnable from frozen features once the head is fitted in closed form on standardised inputs; the gradient-trained probe and bias-only tuning that return a constant are failures of optimisation, not of the features. Moving the encoder, by LoRA, bottleneck adapters or partial fine-tuning alike, adds 3.5 yr and 0.08 AUC on the cohort farther from the pretraining data and nothing measurable on the nearer one. Full fine-tuning at its own learning rate is the most accurate single-task strategy on segmentation and sex classification; its cost is forgetting, which freezing alone prevents: a second segmentation task erased the first under full fine-tuning in either order, EWC reduced the loss on a learnable second task without removing it, and LwF and replay in four forms did no better than fine-tuning.

Three cautions accompany these recommendations. A learning rate shared across strategies favoured the parameter-efficient ones, and pooling cases across seeds produced significance that five runs do not support; both should be checked in any such comparison. A pretrained checkpoint should be validated, tensor by tensor and against a random-initialisation control, rather than assumed to transfer; ours loaded 59 of 82 segmentation tensors and made no measurable difference under LoRA on four tasks. And an evaluation protocol that scores random crops on the model-selection split can inflate results enough to reverse the ordering of methods, which is why we release ours along with the code, the splits and the per-case predictions.

Code and Data Availability

The code, the persisted subject-grouped splits for every task and seed, the per-case predictions behind every table and figure, the table and statistics generators, and the protocol tests are available at https://github.com/ChiShengChen/FewShot3DBrain. The imaging data are obtained from their original sources under the terms listed in Sec. IV-A; no imaging data are redistributed.

Acknowledgment

EJK acknowledges financial support from the National Science and Technology Council (NSTC) of Taiwan under Grant No. NSTC 114-2112-M-A49-036-MY3.

References

[1] FOMO Challenge Organizers, "FOMO-60K: Foundation models for open medical imaging," 2025, FOMO25 Challenge, HuggingFace: FOMO25/FOMO-MRI.

[2] FOMO25, "FOMO baseline codebase," 2025. [Online]. Available: https://github.com/fomo25/baseline-codebase

[3] J. Kirkpatrick, R. Pascanu, N. Rabinowitz *et al.*, "Overcoming catastrophic forgetting in neural networks," *Proc. Natl. Acad. Sci. USA*, vol. 114, no. 13, pp. 3521–3526, 2017.

[4] M. McCloskey and N. J. Cohen, "Catastrophic forgetting in connectionist networks: The sequential learning problem," *Psychol. Learn. Motiv.*, vol. 24, pp. 109–165, 1989.

[5] Z. Li and D. Hoiem, "Learning without forgetting," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 12, pp. 2935–2947, 2018.

[6] E. B. Zaken, S. Ravfogel, and Y. Goldberg, "BitFit: Simple parameter-efficient fine-tuning for transformer-based masked language-models," in *Proc. ACL*, 2022, pp. 1–9.

[7] E. J. Hu, Y. Shen, P. Wallis *et al.*, "LoRA: Low-rank adaptation of large language models," in *Proc. ICLR*, 2022.

[8] J. Y.-C. Hu, M. Su, E.-J. Kuo, Z. Song, and H. Liu, "Computational limits of low-rank adaptation (LoRA) fine-tuning for transformer models," in *Proc. ICLR*, 2025.

[9] A. Hatamizadeh, Y. Tang, V. Nath *et al.*, "UNETR: Transformers for 3D medical image segmentation," in *Proc. WACV*, 2022, pp. 574–584.

[10] H. Wang, S. Guo, J. Ye *et al.*, "SAM-Med3D: Towards general-purpose segmentation models for volumetric medical images," *arXiv preprint arXiv:2310.15161*, 2023.

[11] R. J. Chen, T. Ding, M. Y. Lu *et al.*, "Towards a general-purpose foundation model for computational pathology," *Nat. Med.*, vol. 30, pp. 850–862, 2024.

[12] F. Zenke, B. Poole, and S. Ganguli, "Continual learning through synaptic intelligence," in *Proc. ICML*, 2017, pp. 3987–3995.

[13] S.-A. Rebuffi, A. Kolesnikov, G. Sperl, and C. H. Lampert, "iCaRL: Incremental classifier and representation learning," in *Proc. CVPR*, 2017, pp. 2001–2010.

[14] M. M. Derakhshani, I. Najdenkoska, T. van Sonsbeek *et al.*, "LifeLonger: A benchmark for continual disease classification," in *Proc. MICCAI*, 2022, pp. 314–324.

[15] C. González, A. Ranem, D. Pinto dos Santos, A. Othman, and A. Mukhopadhyay, "Lifelong nnU-Net: A framework for standardized medical continual learning," *Sci. Rep.*, vol. 13, p. 9381, 2023.

[16] N. Houlsby, A. Giurgiu, S. Jastrzebski *et al.*, "Parameter-efficient transfer learning for NLP," in *Proc. ICML*, 2019, pp. 2790–2799.

[17] X. L. Li and P. Liang, "Prefix-tuning: Optimizing continuous prompts for generation," in *Proc. ACL*, 2021, pp. 4582–4597.

[18] A. Mallya and S. Lazebnik, "PackNet: Adding multiple tasks to a single network by iterative pruning," in *Proc. CVPR*, 2018, pp. 7765–7773.

[19] A. Mallya, D. Davis, and S. Lazebnik, "Piggyback: Adapting a single network to multiple tasks by learning to mask weights," in *Proc. ECCV*, 2018, pp. 67–82.

[20] Hugging Face, "PEFT: State-of-the-art parameter-efficient fine-tuning methods," 2023. [Online]. Available: https://github.com/huggingface/peft

[21] U. Baid, S. Ghodasara, S. Mohan *et al.*, "The RSNA-ASNR-MICCAI BraTS 2021 benchmark on brain tumor segmentation and radiogenomic classification," *arXiv preprint arXiv:2107.02314*, 2021.

[22] B. H. Menze, A. Jakab, S. Bauer *et al.*, "The multimodal brain tumor image segmentation benchmark (BRATS)," *IEEE Trans. Med. Imaging*, vol. 34, no. 10, pp. 1993–2024, 2015.

[23] Imperial College London and Guy's Hospital, "IXI dataset," 2008. [Online]. Available: https://brain-development.org/ixi-dataset/

[24] M. R. Hernandez Petzsche, E. de la Rosa, U. Hanning *et al.*, "ISLES 2022: A multi-center magnetic resonance imaging stroke lesion segmentation dataset," *Sci. Data*, vol. 9, no. 1, p. 762, 2022.

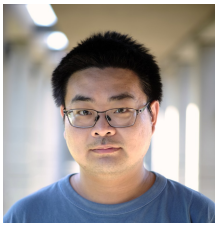
[25] M. Antonelli, A. Reinke, S. Bakas *et al.*, "The medical segmentation decathlon," *Nat. Commun.*, vol. 13, no. 1, p. 4128, 2022.

[26] H. J. Kuijf, J. M. Biesbroek, J. de Bresser *et al.*, "Standardized assessment of automatic segmentation of white matter hyperintensities and results of the WMH segmentation challenge," *IEEE Trans. Med. Imaging*, vol. 38, no. 11, pp. 2556–2568, 2019.

[27] D. S. Marcus, T. H. Wang, J. Parker *et al.*, "Open access series of imaging studies (OASIS): Cross-sectional MRI data in young, middle aged, nondemented, and demented older adults," *J. Cogn. Neurosci.*, vol. 19, no. 9, pp. 1498–1507, 2007.

[28] A. Hatamizadeh, V. Nath, Y. Tang *et al.*, "Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images," in *Proc. BrainLes (MICCAI Workshop)*, 2022, pp. 272–284.

[29] Y. Tang, D. Yang, W. Li *et al.*, "Self-supervised pre-training of Swin transformers for 3D medical image analysis," in *Proc. CVPR*, 2022, pp. 20 730–20 740.



Chi-Sheng Chen received a bachelor degree in Electrophysics from National Chiao Tung University and a master's degree in Biomedical Electronics and Bioinformatics from National Taiwan University. He is currently working at a financial AI startup and also serves as an AI project researcher at Harvard Medical School and Beth Israel Deaconess Medical Center. His research expertise includes time-series analysis, computer vision, brain-computer interfaces (BCI), electroencephalography (EEG), multimodal contrastive learning, and quantum machine learning. Chen has led the development of foundational models for EEG-based image generation, aiming to improve psychiatric treatment and the discovery of biomarkers for drug development. He has designed hybrid quantum-classical architectures for EEG encoding and pioneered quantum deep learning methods for contrastive learning between EEG and visual data. His research also focuses on medical speech processing and the integration of multimodal large language models (LLMs) for clinical and financial application.



Xinyu Zhang received the B.E. and the M.S. degree from the Information and Automatic College, Tianjin University of Science and Technology, Tianjin, China, in 2019 and 2022, separately, where he is currently pursuing the Ph.D. degree in intelligent system engineering with Luddy School of Informatics, Computing, and Engineering, Indiana University Bloomington, Bloomington, IN, USA. His main research includes quantum neural network, deep learning, advanced signal processing, and other disciplines such as information science and large language models (LLMs).



En-Jui Kuo received the B.S. degree in electrophysics from National Yang Ming University, Taipei, Taiwan. He then obtained his M.S. degree in physics from National Taiwan University, Taipei, Taiwan, and completed his Ph.D. in physics at the University of Maryland, USA. Currently, he is an Assistant Professor of department of electrophysics at the National Yang Ming Chiao Tung University, Taiwan.



Rong Fu received the B.S. degree in Communication Engineering from Jiangsu Ocean University, Jiangsu, China, in 2018, and the M.S. degree in Electronic Information Engineering from the College of Information and Automation, Tianjin University of Science and Technology, Tianjin, China, in 2023. She is currently pursuing the Ph.D. degree in Intelligent Systems Engineering with the Luddy School of Informatics, Computing, and Engineering, Indiana University Bloomington, Bloomington, IN, USA. Her research interests include signal processing and deep learning.



Qiuzhe Xie received a bachelor's degree in Electrical Engineering from Hefei University of Technology and a master's degree in Electronics Engineering from National Cheng Kung University, followed by a Ph.D. in Electronics Engineering from National Taiwan University. His research expertise encompasses biosensors, advanced semiconductor processes, nanomaterials, and microfabrication techniques. He has spearheaded the development of nano-structural electrode arrays with ultra-thin dielectric stacking and leveraged quantum computing algorithms to enhance sensing precision. Furthermore, he has engineered CRISPR-based electrochemical sensors and developed innovative water purification systems for semiconductor manufacturing. Extending his work to practical application, he also focuses on the commercialization of medical devices, having validated business models for cardiosensor technologies through the NSF program.



Fan Zhang received the B.S. degree in Statistics from Beijing Normal University, Beijing, China, in 2016, the M.S. degree in statistics from National Cheng Kung University, Taiwan, in 2018, and the Ph.D. degree in statistics from Arizona State University, Tempe, AZ, USA, in 2024. She is currently an Assistant Professor with the Department of Mathematics, Boise State University, Boise, ID, USA. Her research interests include design and analysis of experiments, variable selection, uncertainty quantification, Bayesian modelling and inference, and quantum-enhanced machine learning.